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## Shock absorber


### Abstract

This shock absorber includes a first damping force generation mechanism provided in a passage of a piston, and a second damping force generation mechanism provided in a piston rod. The second damping force generation mechanism includes a valve seat member, a sub-valve provided in the valve seat member, and a cap member covering one end side of the second damping force generation mechanism and at least a part of an outer circumference of the valve seat member. A communication passage is formed on one end side of the cap member. A biasing member provided so that one end surface side is in contact with an outer circumferential side of the cap member with respect to the communication passage, and a bendable flexible disc provided so that the other end surface side of the biasing member is in contact therewith.



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## Background/Summary

(1) The present invention relates to a shock absorber.
(2) Priority is claimed on Japanese Patent Application No. 2020-128095 filed on Jul. 29, 2020, the content of which is incorporated herein by reference.
BACKGROUND ART
(3) In shock absorbers, there are those having two valves that open in the same stroke (see, for example, Patent Document 1).
CITATION LIST
Patent Document
(4) [Patent Document 1]
(5) Japanese Examined Patent Application, Second Publication No. H2-41666
SUMMARY OF INVENTION
Technical Problem
(6) When two valves that open in the same stroke are provided, one valve can be made to open in a region in which a piston speed is lower than that of the other valve, and both valves can be made to open in a region in which the piston speed is higher than that. In such a structure, there is a likelihood that abnormal noise will be generated especially at the time of a high-frequency input.

(7) The present invention provides a shock absorber capable of suppressing generation of abnormal noise.
Solution to Problem
(8) According to a first aspect of the present invention, a shock absorber includes a cylinder in which a working fluid is sealed, a piston provided to be slidable in the cylinder and partitioning the inside of the cylinder into two chambers, a piston rod connected to the piston and extending to the outside of the cylinder, a passage through which a working fluid flows from the chamber on an upstream side to the chamber on a downstream side in the cylinder due to movement of the piston, a first damping force generation mechanism provided in the passage formed in the piston to generate a damping force, and a second damping force generation mechanism disposed separately from the first damping force generation mechanism and provided in the piston rod. The second damping force generation mechanism includes a valve seat member, a sub-valve provided in the valve seat member, and a cap member covering one end side of the second damping force generation mechanism and at least a part of an outer circumference of the valve seat member. A communication passage which allows the inside and outside of the cap member to communicate is formed at one end side of the cap member. A biasing member provided so that one end surface side is in contact with an outer circumferential side of the cap member with respect to the communication passage, and a bendable flexible disc provided so that the other end surface side of the biasing member is in contact therewith are disposed.
Advantageous Effects of Invention
(9) According to the shock absorber described above, generation of abnormal noise can be suppressed.

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## Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a cross-sectional view illustrating a shock absorber according to a first embodiment of the present invention.
(2) FIG. **2** is a partial cross-sectional view illustrating the vicinity of a piston of the shock absorber according to the first embodiment of the present invention.
(3) FIG. **3** is a partial cross-sectional view illustrating a main part of the shock absorber according to the first embodiment of the present invention.
(4) FIG. **4** is a partial cross-sectional view illustrating a main part of a shock absorber according to a second embodiment of the present invention.
(5) FIG. **5** is a partial cross-sectional view illustrating a main part of a shock absorber according to a third embodiment of the present invention.
(6) FIG. **6** is a plan view illustrating a planar disc of the shock absorber according to the third embodiment of the present invention.
(7) FIG. **7** is a partial cross-sectional view illustrating a main part of a shock absorber according to a fourth embodiment of the present invention.
(8) FIG. **8** is a plan view illustrating an outer disc of the shock absorber according to the fourth embodiment of the present invention.
(9) FIG. **9** is a partial cross-sectional view illustrating a main part of a shock absorber according to a fifth embodiment of the present invention.
(10) FIG. **10** is a plan view illustrating a guide member of the shock absorber according to the fifth embodiment of the present invention.
(11) FIG. **11** is a partial cross-sectional view illustrating a main part of a shock absorber according to a sixth embodiment of the present invention.
DESCRIPTION OF EMBODIMENTS

First Embodiment
(12) A first embodiment will be described on the basis of FIGS. **1** to **3**. Further, in the following, for convenience of explanation, an upper side in FIGS. **1** to **5**, FIG. **7**, FIG. **9**, and FIG. **11** will be referred to with “upper” and a lower side in FIGS. **1** to **5**, FIG. **7**, FIG. **9**, and FIG. **11** will be referred to with “lower.”
(13) <Configuration>
(14) A shock absorber **1** of the first embodiment is a shock absorber used in a suspension device of a railway vehicle or an automobile such as a two-wheeled vehicle, a four-wheeled vehicle, or the like. Specifically, a shock absorber used in a suspension device of a four-wheeled vehicle will be described. As illustrated in FIG. **1**, the shock absorber **1** is a shock absorber of a dual-tube type including a cylinder **4** having a cylindrical inner tube **2** and a bottomed cylindrical outer tube **3** which has a diameter larger than that of the inner tube **2** and is provided on an outer side of the inner tube **2** in a radial direction. A space between the outer tube **3** and the inner tube **2** is a reservoir chamber **5**.
(15) The outer tube **3** includes a barrel member **8** having a stepped cylindrical shape in which both axial end sides are smaller than an axial intermediate part in diameter, and a bottom member **9** that closes one end portion of the barrel member **8** in the axial direction. A side of the barrel member **8** opposite to the bottom member **9** is an opening.
(16) The shock absorber **1** includes an annular valve body **12** provided at one end portion of the inner tube **2** in the axial direction, and an annular rod guide **13** provided at the other end portions of the inner tube **2** and the outer tube **3** in the axial direction. The valve body **12** constitutes a base valve **15**. An outer circumferential portion of the base valve **15** has a stepped shape. An outer circumferential portion of the rod guide **13** also has a stepped shape. A large diameter portion of the rod guide **13** is fitted to the barrel member **8**.
(17) One end portion of the inner tube **2** in the axial direction is fitted to a small diameter portion of an outer circumferential portion of the valve body **12**. The inner tube **2** is engaged with the bottom member **9** of the outer tube **3** via the valve body **12**. The other end portion of the inner tube **2** in the axial direction is fitted to a small diameter portion of an outer circumferential portion of the rod guide **13**. The inner tube **2** is engaged with the barrel member **8** of the outer tube **3** via the rod guide **13**. In this state, the inner tube **2** is radially positioned with respect to the outer tube **3**. Between the valve body **12** and the bottom member **9**, the inner tube **2** and the outer tube **3** communicate with each other via a passage groove **16** formed in the valve body **12**, thereby constituting the reservoir chamber **5** similarly to a space between the inner tube **2** and the outer tube **3**.
(18) The shock absorber **1** includes an annular seal member **18** on a side of the rod guide **13** opposite to the bottom member **9**. The seal member **18** is also fitted to an inner circumferential portion of the barrel member **8** similarly to the rod guide **13**. A locking part **19** is formed at an end portion of the barrel member **8** opposite to the bottom member **9** by plastically deforming the barrel member **8** inward in the radial direction by swaging processing such as curling processing. The seal member **18** is sandwiched between the locking part **19** and the rod guide **13**. The seal member **18** closes the opening of the outer tube **3**. Specifically, the seal member **18** is an oil seal.
(19) The shock absorber **1** includes a piston **21** provided in the cylinder **4**. The piston **21** is provided to be slidable in the inner tube **2** of the cylinder **4**. The piston **21** partitions the inside of the inner tube **2** into two chambers including an upper chamber **22** and a lower chamber **23**. The upper chamber **22** is provided between the piston **21** and the rod guide **13** inside the inner tube **2**. The lower chamber **23** is provided between the piston **21** and the valve body **12** in the inner tube **2**. The lower chamber **23** is divided from the reservoir chamber **5** by the valve body **12**. In the cylinder **4**, an oil fluid L is sealed as a working fluid in the upper chamber **22** and the lower chamber **23**. In the reservoir chamber **5**, a gas G and the oil fluid L are sealed as working fluids.
(20) The shock absorber **1** includes a piston rod **25**. In the piston rod **25**, one end side portion in the

axial direction is disposed inside the cylinder **4** to be connected and fixed to the piston **21**, and the other end side portion extends to the outside of the cylinder **4**. The piston rod **25** is made of a metal. The piston rod **25** passes through the upper chamber **22** and does not pass through the lower chamber **23**. Therefore, the upper chamber **22** is a rod side chamber through which the piston rod **25** passes, and the lower chamber **23** is a bottom side chamber on the bottom member **9** side of the cylinder **4**.
(21) The piston **21** and the piston rod **25** move together. The piston **21** moves to the upper chamber **22** side in an extension stroke of the shock absorber **1** in which a protrusion amount of the piston rod **25** from the cylinder **4** increases. The piston **21** moves to the lower chamber **23** side in a compression stroke in which the protrusion amount of the piston rod **25** from the cylinder **4** decreases.
(22) Both the rod guide **13** and the seal member **18** are annular. The piston rod **25** is slidably inserted through the inside of the rod guide **13** and the seal member **18** and extends from the inside of the cylinder **4** to the outside. One end side portion of the piston rod **25** in the axial direction is fixed to the piston **21** inside the cylinder **4**. The other end side portion of the piston rod **25** in the axial direction protrudes to the outside of the cylinder **4** via the rod guide **13** and the seal member **18**.
(23) The rod guide **13** guides movement of the piston rod **25** by supporting the piston rod **25** to be movable with respect to the cylinder **4** in the axial direction while restricting its movement in the radial direction. An outer circumferential portion of the seal member **18** is in close contact with the cylinder **4**. An inner circumferential portion of the seal member **18** is in sliding contact with an outer circumferential portion of the piston rod **25** that moves in the axial direction. Thereby, the seal member **18** prevents the oil fluid L and the gas G in the cylinder **4** from leaking to the outside.
(24) The piston rod **25** includes a main shaft part **27** and a mounting shaft part **28** which has a diameter smaller than that of the main shaft part **27**. The main shaft part **27** is slidably fitted to the rod guide **13** and the seal member **18**. The mounting shaft part **28** is disposed in the cylinder **4** and connected to the piston **21** and the like. An end portion of the main shaft part **27** on the mounting shaft part **28** side is a shaft step part **29** which extends in a direction perpendicular to the axis.
(25) On an outer circumferential portion of the mounting shaft part **28**, a passage notch part **30** extending in the axial direction is formed at an intermediate position in the axial direction, and a male screw **31** is formed at a distal end position on a side opposite to the main shaft part **27** in the axial direction. The passage notch part **30** is formed, for example, by cutting the outer circumferential portion of the mounting shaft part **28** into a plane shape on a plane parallel to a central axis of the mounting shaft part **28**. The passage notch part **30** can be formed in a so-called width across flat shape at two positions on the mounting shaft part **28** that differ by 180 degrees in the circumferential direction.
(26) In the shock absorber **1**, for example, the protruding portion of the piston rod **25** from the cylinder **4** is disposed on an upper portion to be supported by a vehicle body, and the bottom member **9** of the cylinder **4** is disposed on a lower portion to be connected to a wheel side.
(27) Conversely, the cylinder **4** side may be supported by the vehicle body, and the piston rod **25** may be connected to the wheel side.
(28) As illustrated in FIG. **2**, the piston **21** is constituted by a piston main body **36** made of a metal and connected in contact with the piston rod **25**, and an annular slide member **37** made of a synthetic resin and integrally mounted on an outer circumferential surface of the piston main body **36** to slide the inside of the inner tube **2** of the cylinder **4**.
(29) The piston main body **36** includes a plurality (only one is illustrated in FIG. **2** because it is a cross section) of passage holes **38** that allow communication between the upper chamber **22** and the lower chamber **23**, and a plurality (only one is illustrated in FIG. **2** because it is a cross section) of passage holes **39** that allow communication between the upper chamber **22** and the lower chamber **23**.

(30) The plurality of passage holes **38** are formed with a uniform pitch with the passage holes **39** interposed therebetween in a circumferential direction of the piston main body **36** and constitute one half of the passage holes **38** and **39** in number. The plurality of passage holes **38** have a crank shape having two bending points. The plurality of passage holes **38** on the lower chamber **23** side of the piston **21** in the axial direction open further inward in the radial direction of the piston **21** than those on the upper chamber **22** side. An annular groove **55** having an annular shape that allows the plurality of passage holes **38** to communicate with each other is formed in the piston main body **36** on the lower chamber **23** side in the axial direction.
(31) A first damping force generation mechanism **41** is provided on the lower chamber **23** side of the annular groove **55**. The first damping force generation mechanism **41** opens and closes passages in the annular groove **55** and the plurality of passage holes **38** to generate a damping force. When the first damping force generation mechanism **41** is disposed on the lower chamber **23** side, passages in the plurality of passage holes **38** and the annular groove **55** serve as extension-side passages through which the oil fluid L flows from the upper chamber **22** on an upstream side toward the lower chamber **23** on a downstream side when the piston **21** moves to the upper chamber **22** side, that is, during an extension stroke. The first damping force generation mechanism **41** provided with respect to these passages in the plurality of passage holes **38** and the annular groove **55** serves as an extension-side damping force generation mechanism that generates a damping force by suppressing a flow of the oil fluid L on the extension side from the passages in the plurality of passage holes **38** and the annular groove **55** to the lower chamber **23**.
(32) The passage holes **39** constituting the remaining half of the passage holes **38** and **39** are formed with a uniform pitch with the passage holes **38** interposed therebetween in the circumferential direction of the piston main body **36**. The plurality of passage holes **39** have a crank shape having two bending points. The plurality of passage holes **39** on the upper chamber **22** side of the piston **21** in the axial direction open further inward in the radial direction of the piston **21** than those on the lower chamber **23** side. An annular groove **56** having an annular shape that allows the plurality of passage holes **39** to communicate with the upper chamber **22** side in the axial direction is formed in the piston main body **36**.
(33) A first damping force generation mechanism **42** that opens and closes passages in the plurality of passage holes **39** and the annular groove **56** to generate a damping force is provided on the upper chamber **22** side of the annular groove **56**. When the first damping force generation mechanism **42** is disposed on the upper chamber **22** side, the passages in the plurality of passage holes **39** and the annular groove **56** serve as compression-side passages through which the oil fluid L flows from the lower chamber **23** on an upstream side toward the upper chamber **22** on a downstream side when the piston **21** moves to the lower chamber **23** side, that is, during a compression stroke. The first damping force generation mechanism **42** provided with respect to these passages in the plurality of passage holes **39** and the annular groove **56** serves as a compression-side damping force generation mechanism that generates a damping force by suppressing a flow of the oil fluid L on the compression side from the passages in the plurality of passage holes **39** and the annular groove **56** to the upper chamber **22**.
(34) The piston main body **36** has substantially a disc shape. At a center of the piston main body **36** in the radial direction, an insertion hole **44** into which the mounting shaft part **28** of the piston rod **25** is inserted is formed to penetrate in the axial direction. The insertion hole **44** includes a small diameter hole portion **45** on one side in the axial direction into which the mounting shaft part **28** of the piston rod **25** is fitted, and a large diameter hole portion **46** on the other side in the axial direction having a larger diameter than the small diameter hole portion **45**. The small diameter hole portion **45** is provided on the upper chamber **22** side in the axial direction, and the large diameter hole portion **46** is provided on the lower chamber **23** side in the axial direction. When the mounting shaft part **28** is fitted into the small diameter hole portion **45**, the piston **21** is radially positioned with respect to the piston rod **25**.

(35) At an end portion of the piston main body **36** on the lower chamber **23** side in the axial direction, an annular inner seat part **47** is formed on an inner side in the radial direction of the piston main body **36** with respect to the opening of the annular groove **55** on the lower chamber **23** side. At an end portion of the piston main body **36** on the lower chamber **23** side in the axial direction, an annular valve seat part **48** constituting a part of the first damping force generation mechanism **41** is formed on an outer side in the radial direction of the piston main body **36** with respect to the opening of the annular groove **55** on the lower chamber **23** side.
(36) At an end portion of the piston main body **36** on the upper chamber **22** side in the axial direction, an annular inner seat part **49** is formed on an inner side in the radial direction of the piston main body **36** with respect to an opening of the annular groove **56** on the upper chamber **22** side. At an end portion of the piston main body **36** on the upper chamber **22** side in the axial direction, an annular valve seat part **50** constituting a part of the first damping force generation mechanism **42** is formed on an outer side in the radial direction of the piston main body **36** with respect to the opening of the annular groove **56** on the upper chamber **22** side.
(37) In the insertion hole **44** of the piston main body **36**, the large diameter hole portion **46** is provided on the inner seat part **47** side in the axial direction with respect to the small diameter hole portion **45**. A passage in the large diameter hole portion **46** of the piston main body **36** and a piston rod passage part **51** in the passage notch part **30** of the piston rod **25** overlap in axial position and are in constant communication with each other.
(38) In the piston main body **36**, an outer side of the valve seat part **48** in the radial direction has a stepped shape whose height in the axial direction is lower than that of the valve seat part **48**. The openings of the compression-side passage holes **39** on the lower chamber **23** side are disposed in the portion with the stepped shape. Similarly, in the piston main body **36**, an outer side of the valve seat part **50** in the radial direction has a stepped shape whose height in the axial direction is lower than that of the valve seat part **50**. The openings of the extension-side passage holes **38** on the upper chamber **22** side are disposed in the portion with the stepped shape.
(39) The compression-side first damping force generation mechanism **42** includes the valve seat part **50** of the piston **21**. The first damping force generation mechanism **42** includes one disc **63**, a plurality of (specifically, two) discs **64** having the same inner diameter and the same outer diameter, a plurality of (specifically, three) discs **65** having the same inner diameter and the same outer diameter, a plurality of (specifically, two) discs **66** having the same inner diameter and the same outer diameter, one disc **67**, one disc **68**, and one annular member **69** in that order from the piston **21** side in the axial direction. The discs **63** to **68** and the annular member **69** are made of a metal and each have a bored disc shape with a constant thickness. The discs **63** to **68** and the annular member **69** are all radially positioned with respect to the piston rod **25** with the mounting shaft part **28** fitted therein. The discs **63** to **68** are plane discs (planar discs without a protrusion protruding in the axial direction).
(40) The disc **63** has an outer diameter larger than an outer diameter of the inner seat part **49** of the piston **21** and smaller than an inner diameter of the valve seat part **50**. The disc **63** is in constant contact with the inner seat part **49**. The plurality of discs **64** have an outer diameter equal to an outer diameter of the valve seat part **50** of the piston **21**. The plurality of discs **64** can be seated on the valve seat part **50**. The plurality of discs **65** have an outer diameter smaller than an outer diameter of the discs **64**. The plurality of discs **66** have an outer diameter smaller than the outer diameter of the discs **65**. The disc **67** has an outer diameter smaller than the outer diameter of the discs **66** and equal to the outer diameter of the inner seat part **49** of the piston **21**. The disc **68** has an outer diameter equal to the outer diameter of the discs **65**. The annular member **69** has an outer diameter smaller than the outer diameter of the disc **68** and larger than an outer diameter of the shaft step part **29** of the piston rod **25**. The annular member **69** has a larger thickness and higher rigidity than the discs **63** to **68**. The annular member **69** is in contact with the shaft step part **29**.
(41) The plurality of discs **64**, the plurality of discs **65**, and the plurality of discs **66** constitute a

compression-side main valve **71** that can be separated from and seated on the valve seat part **50**. When the main valve **71** is separated from the valve seat part **50**, the passages in the plurality of passage holes **39** and the annular groove **56** are allowed to communicate with the upper chamber **22**, and a flow of the oil fluid L between the main valve **71** and the valve seat part **50** is suppressed to generate a damping force. The annular member **69**, together with the disc **68**, restricts deformation of the main valve **71** in an opening direction beyond a specified limit by coming into contact with the main valve **71**.

(42) The passages in the plurality of passage holes **39** and the annular groove **56** and the passage between the main valve **71** and the valve seat part **50** that appears when the valve opens are formed in the piston **21**, and constitute a compression-side first passage **72** in which the oil fluid L flows from the lower chamber **23** on the upstream side to the upper chamber **22** on the downstream side in the cylinder **4** due to movement of the piston **21** to the lower chamber **23** side. The compression-side first damping force generation mechanism **42** generating a damping force includes the main valve **71** and the valve seat part **50**. Therefore, the first damping force generation mechanism **42** is provided in the first passage **72**. The first passage **72** is formed in the piston **21** that includes the valve seat part **50**. The oil fluid L passes through the first passage **72** when the piston rod **25** and the piston **21** move to the compression side.

(43) In the compression-side first damping force generation mechanism **42**, a fixed orifice that allows communication between the upper chamber **22** and the lower chamber **23** even in a state in which the valve seat part **50** and the main valve **71** that is in contact with the valve seat part **50** are in contact with each other is not formed in either of the valve seat part **50** and the main valve **71**. That is, the compression-side first damping force generation mechanism **42** does not allow communication between the upper chamber **22** and the lower chamber **23** if the valve seat part **50** and the main valve **71** are in contact with each other over the entire circumference. In other words, the first passage **72** does not include a fixed orifice formed to allow constant communication between the upper chamber **22** and the lower chamber **23** and is not a passage that allows constant communication between the upper chamber **22** and the lower chamber **23**.

(44) The extension-side first damping force generation mechanism **41** includes the valve seat part **48** of the piston **21**. The first damping force generation mechanism **41** includes one disc **82**, one disc **83**, a plurality of (specifically, four) discs **84** having the same inner diameter and the same outer diameter, one disc **85**, a plurality of (specifically three) discs **86** having the same inner diameter and the same outer diameter, and one disc **87** in that order from the piston **21** side in the axial direction. The discs **82** to **87** are plane discs made of a metal each having a bored disc shape with a constant thickness. The discs **82** to **87** are all radially positioned with respect to the piston rod **25** with the mounting shaft part **28** fitted therein.

(45) The disc **82** has an outer diameter larger than an outer diameter of the inner seat part **47** of the piston **21** and smaller than an inner diameter of the valve seat part **48**. The disc **82** is in constant contact with the inner seat part **47**. As illustrated in FIG. **3**, a notch part **90** that allows the passages in the annular groove **55** and the plurality of passage holes **38** to constantly communicate with the passage in the large diameter hole portion **46** of the piston **21** and the piston rod passage part **51** in the passage notch part **30** of the piston rod **25** is formed in the disc **82** from an intermediate position on an outer side of the inner seat part **47** in the radial direction to an inner circumferential edge portion thereof. The notch part **90** is formed at the time of press-forming the disc **82**. The notch part **90** is adjacent to and faces the large diameter hole portion **46** of the piston **21**. The disc **83** has the same outer diameter as the disc **82** but a notch part as in the disc **82** is not formed therein. The plurality of discs **84** have an outer diameter equal to an outer diameter of the valve seat part **48** of the piston **21**. The plurality of discs **84** can be seated on the valve seat part **48**. The disc **85** has an outer diameter smaller than the outer diameter of the discs **84**. The plurality of discs **86** have an outer diameter smaller than the outer diameter of the disc **85**. The disc **87** has an outer diameter smaller than the outer diameter of the discs **86** and slightly larger than the outer diameter

of the inner seat part **47** of the piston **21**.
(46) The plurality of discs **84**, the one disc **85**, and the plurality of discs **86** constitute an extension-side main valve **91** that can be separated from and seated on the valve seat part **48**. When the main valve **91** is separated from the valve seat part **48**, the passages in the annular groove **55** and the plurality of passage holes **38** are allowed to communicate with the lower chamber **23**, and a flow of the oil fluid L between the main valve **91** and the valve seat part **48** is suppressed to generate a damping force.
(47) As illustrated in FIG. **2**, the passages in the plurality of passage holes **38** and the annular groove **55** and a passage between the main valve **91** and the valve seat part **48** that appears when the valve opens are formed in the piston **21**, and constitute an extension-side first passage **92** in which the oil fluid L flows from the upper chamber **22** on the upstream side to the lower chamber **23** on the downstream side in the cylinder **4** due to movement of the piston **21** to the upper chamber **22** side. The extension-side first damping force generation mechanism **41** generating a damping force includes the main valve **91** and the valve seat part **48**. Therefore, the first damping force generation mechanism **41** is provided in the first passage **92**. The first passage **92** is formed in the piston **21** that includes the valve seat part **48**. The oil fluid L passes through the first passage **92** when the piston rod **25** and the piston **21** move to the extension side.
(48) In the extension-side first damping force generation mechanism **41**, a fixed orifice that allows communication between the upper chamber **22** and the lower chamber **23** even in a state in which the valve seat part **48** and the main valve **91** that is in contact with the valve seat part **48** are in contact with each other is not formed in either of the valve seat part **48** and the main valve **91**. That is, the extension-side first damping force generation mechanism **41** does not allow communication between the upper chamber **22** and the lower chamber **23** if the valve seat part **48** and the main valve **91** are in contact with each other over the entire circumference. In other words, the first passage **92** does not have a fixed orifice formed to allow constant communication between the upper chamber **22** and the lower chamber **23** and is not a passage that allows constant communication between the upper chamber **22** and the lower chamber **23**.
(49) As illustrated in FIG. **3**, on a side of the extension-side first damping force generation mechanism **41** opposite to the piston **21**, one cap member main body **95**, one disc **97**, a plurality of (specifically, two) discs **98** having the same inner diameter and the same outer diameter, one disc **99**, one flexible disc **100**, one disc **101**, a plurality of (specifically, two) discs **102** having the same inner diameter and the same outer diameter, one disc **103**, one spring contact disc **104**, one spring member **105**, one disc **106**, one sub-valve **107**, one valve seat member **109** in which one O-ring **108** is provided on an outer circumferential side, one sub-valve **110**, one disc **111**, one spring member **112**, one disc **113**, and one annular member **114** are provided in that order from the first damping force generation mechanism **41** side with the mounting shaft part **28** of the piston rod **25** fitted to inner sides thereof. The cap member main body **95**, the discs **97** to **99**, the flexible disc **100**, the discs **101** to **103**, the spring contact disc **104**, the spring member **105**, the disc **106**, the sub-valve **107**, the valve seat member **109**, the sub-valve **110**, the disc **111**, the spring member **112**, the disc **113**, and the annular member **114** are radially positioned with respect to the piston rod **25** when the mounting shaft part **28** is fitted to inner sides thereof.
(50) An annular disc spring **116** (biasing member) is provided on an outer side of the disc **97**, the discs **98**, and the disc **99** in the radial direction. An annular disc spring **117** (biasing member) is provided on an outer side of the disc **101**, the discs **102**, and the disc **103** in the radial direction.
(51) As illustrated in FIG. **2**, the male screw **31** is formed on the mounting shaft part **28** of the piston rod **25** at a portion protruding further than the annular member **114**. A nut **119** is screwed onto the male screw **31**. The nut **119** is in contact with the annular member **114**.
(52) The annular member **69**, the discs **63** to **68**, the piston **21**, the discs **82** to **87**, the cap member main body **95**, and the discs **97** to **99** illustrated in FIG. **3**, the flexible disc **100**, the discs **101** to **103**, the spring contact disc **104**, the spring member **105**, the disc **106**, the sub-valve **107**, the valve

seat member **109**, the sub-valve **110**, the disc **111**, the spring member **112**, the disc **113**, and the annular member **114** are axially clamped at least at their radially inner circumferential sides by the shaft step part **29** of the piston rod and the nut **119** and fixed to the piston rod **25** as illustrated in FIG. **2**. In this state, as illustrated in FIG. **3**, the discs **97** to **99**, the flexible disc **100**, the discs **101** to **103**, the spring contact disc **104**, the spring member **105**, the disc **106**, the sub-valve **107**, the valve seat member **109**, the sub-valve **110**, the disc **111**, the spring member **112**, and the disc **113** are disposed in the cap member main body **95**.
(53) The cap member main body **95**, the discs **97** to **99**, **101** to **103**, **106**, **111**, and **113**, the flexible disc **100**, the spring contact disc **104**, the spring members **105** and **112**, the sub-valves **107** and **110**, the valve seat member **109**, and the annular member **114**, and the disc springs **116** and **117** are all made of metals. The discs **97** to **99**, **101** to **103**, **106**, **111**, and **113**, the flexible disc **100**, the spring contact disc **104**, the sub-valves **107** and **110**, and the annular member **114** are all plane discs having a bored disc shape with a constant thickness. The cap member main body **95**, the valve seat member **109**, and the spring members **105** and **112** are annular.
(54) The cap member main body **95** is an integrally formed product having a bottomed cylindrical shape. The cap member main body **95** is formed by, for example, plastic working or cutting work of a metal plate. The cap member main body **95** includes a bottom part **122** having a bored disc shape with a constant thickness, an intermediate curved part **123** extending from an outer circumferential edge portion of the bottom part **122** to one side of the bottom part **122** in the axial direction while a diameter thereof increases, and a cylindrical part **124** having a cylindrical shape extending in a direction opposite to the bottom part **122** from an end edge portion of the intermediate curved part **123** on a side opposite to the bottom part **122**.
(55) The bottom part **122** has a bored disc shape with a constant width in the radial direction over the entire circumference. The mounting shaft part **28** of the piston rod **25** is fitted to an inner circumferential portion of the bottom part **122**. When the mounting shaft part **28** is fitted to the inner circumferential portion of the bottom part **122**, the cap member main body **95** is radially positioned with respect to the piston rod **25** and coaxially disposed therewith. Between the inner circumferential portion and the outer circumferential portion of the bottom part **122**, a plurality of passage holes **126** are formed to penetrate the bottom part **122** in an axial direction of the bottom part **122**. The plurality of passage holes **126** are disposed at regular intervals in a circumferential direction of the bottom part **122** at positions equidistant from a center of the bottom part **122**. In the cap member main body **95**, the bottom part **122** is disposed to be positioned on the piston **21** side with respect to the cylindrical part **124** and is in contact with the disc **87**. The cap member main body **95** is fitted onto the mounting shaft part **28** at the inner circumferential portion of the bottom part **122**. An outer diameter of the disc **87** is smaller than twice a shortest distance connecting a radial center of the cap member main body **95** and the passage hole **126**.
(56) The intermediate curved part **123** has an annular shape that is coaxial with the bottom part **122**. A cross section of the intermediate curved part **123** in a plane including a central axis thereof has a curved shape that is convex outward in the radial direction and toward the bottom part **122** side in the axial direction. The cylindrical part **124** is also coaxial with the bottom part **122** and the intermediate curved part **123**.
(57) The cap member main body **95** has a thickness larger than a thickness of one of the discs **84** to **86** in addition to having a bottomed cylindrical shape, and therefore has higher rigidity than the discs **84** to **86**. Therefore, the cap member main body **95** restricts deformation of the main valve **91**, which is constituted by the plurality of discs **84** to **86**, in an opening direction beyond a specified limit by coming into contact with the main valve **91**.
(58) The disc **97** has a constant width in the radial direction over the entire circumference. An outer diameter of the disc **97** is smaller than twice a shortest distance connecting the radial center of the cap member main body **95** and the passage hole **126**. The disc **97** has the same thickness as the bottom part **122** of the cap member main body **95**.

(59) The discs **98** have a constant width in the radial direction over the entire circumference. An outer diameter of the disc **98** is larger than an outer diameter of the disc **97** and slightly larger than twice a shortest distance connecting the radial center of the cap member main body **95** and the passage hole **126**. A thickness of one of the discs **98** is smaller than a thickness of the disc **97**.
(60) The disc **99** has a constant width in the radial direction over the entire circumference. An outer diameter of the disc **99** is smaller than the outer diameter of the disc **97**. The disc **99** has a thickness equivalent to the thickness of one of the discs **98**.
(61) The flexible disc **100** has a constant width in the radial direction over the entire circumference. An outer diameter of the flexible disc **100** is larger than twice a longest distance connecting the radial center of the cap member main body **95** and the passage hole **126**, and is slightly smaller than an inner diameter of the cylindrical part **124** of the cap member main body **95**. The flexible disc **100** has a thickness equivalent to a thickness of the disc **99**.
(62) The disc **101** is a common part having the same shape as the disc **99**. The disc **102** is a common part having the same shape as the disc **98**. The disc **103** is a common part having the same shape as the disc **97**.
(63) The spring contact disc **104** has a constant width in the radial direction over the entire circumference. An outer diameter of the spring contact disc **104** is slightly smaller than the inner diameter of the cylindrical part **124** of the cap member main body **95**. Therefore, there is a gap between the spring contact disc **104** and the cylindrical part **124** of the cap member main body **95**. A passage hole **301** is formed to penetrate the spring contact disc **104** in the axial direction so that a position of the spring contact disc **104** in the radial direction is substantially coincident with the passage hole **126** of the cap member main body **95**.
(64) The inside of the passage hole **301** is a communication passage **302**. The spring contact disc **104**, together with the cap member main body **95**, constitutes a cap member **305**. The bottom part **122** of the cap member main body **95** constitutes one bottom part of the cap member **305**, and the spring contact disc **104** constitutes the other bottom part of the cap member **305**.
(65) The disc spring **116** is provided between the bottom part **122** of the cap member main body **95** and the flexible disc **100**. The disc spring **116** is formed by press forming a single plate material. The disc spring **116** has a bored disc shape in which a hole **310** is formed at a center in the radial direction and a width in the radial direction is constant over the entire circumference. The discs **97** to **99** are disposed on an inner side of the disc spring **116** in the radial direction with a gap with respect to an inner circumferential edge portion of disc spring **116** in the radial direction. The cylindrical part **124** of the cap member main body **95** is disposed on an outer side of the disc spring **116** in the radial direction. The disc spring **116** includes an inner tapered plate part **311**, an inner curved plate part **312**, an intermediate tapered plate part **313**, an outer curved plate part **314**, and an outer tapered plate part **315** in order from an inner circumferential edge portion side toward an outer circumferential edge portion side thereof.
(66) The inner tapered plate part **311** has a constant width in the radial direction over the entire circumference. An inner circumferential edge portion of the inner tapered plate part **311** is the inner circumferential edge portion of the disc spring **116**. An inner side of the inner circumferential edge portion is the hole **310**. The inner tapered plate part **311** has a tapered shape that extends outward in the radial direction from the inner circumferential edge portion thereof to be inclined such that it is positioned further toward the bottom part **122** side in the axial direction toward the outside in the radial direction. An inner diameter of the inner tapered plate part **311**, that is, an inner diameter of the hole **310**, is larger than the outer diameter of the disc **98**.
(67) The intermediate tapered plate part **313** has a constant width in the radial direction over the entire circumference. The intermediate tapered plate part **313** has a tapered shape that extends outward in the radial direction from an outer circumferential side of the inner tapered plate part **311** to be inclined such that it is positioned further toward the flexible disc **100** side in the axial direction toward the outside in the radial direction. The intermediate tapered plate part **313** has a

radial width larger than a radial width of the inner tapered plate part **311**. The intermediate tapered plate part **313** has an axial length larger than an axial length of the inner tapered plate part **311**.
(68) The outer tapered plate part **315** has a constant width in the radial direction over the entire circumference. The outer tapered plate part **315** has a tapered shape that extends outward in the radial direction from an outer circumferential side of the intermediate tapered plate part **313** to be inclined such that it is positioned further toward the bottom part **122** side in the axial direction toward the outside in the radial direction. The outer tapered plate part **315** has a radial width smaller than the radial width of the intermediate tapered plate part **313** and equal to the radial width of the inner tapered plate part **311**. The outer tapered plate part **315** has an axial length smaller than the axial length of the intermediate tapered plate part **313** and equal to the axial length of the inner tapered plate part **311**. An outer circumferential edge portion of the outer tapered plate part **315** is an outer circumferential edge portion of the disc spring **116**. An outer diameter of the disc spring **116** is slightly smaller than the inner diameter of the cylindrical part **124** of the cap member main body **95**.
(69) The inner curved plate part **312** is formed between the inner tapered plate part **311** and the intermediate tapered plate part **313** by bending the intermediate tapered plate part **313** with respect to the inner tapered plate part **311**. The inner curved plate part **312** connects the outer circumferential side of the inner tapered plate part **311** and an inner circumferential side of the intermediate tapered plate part **313**. A cross section of the inner curved plate part **312** in a plane including a central axis of the disc spring **116** has an arcuate shape that is convex toward the bottom part **122** side. The inner curved plate part **312** has a circular shape centered on the central axis of the disc spring **116**. The inner curved plate part **312** is disposed on the same plane perpendicular to the center axis of the disc spring **116** over the entire circumference.
(70) The disc spring **116** is in contact with the bottom part **122** at the inner curved plate part **312**. The inner curved plate part **312** is in contact with the bottom part **122** at a position on a radially outer side of all the passage holes **126** of the bottom part **122** over the entire circumference.
(71) The outer curved plate part **314** is formed between the intermediate tapered plate part **313** and the outer tapered plate part **315** by bending the outer tapered plate part **315** with respect to the intermediate tapered plate part **313**. The outer curved plate part **314** connects the outer circumferential side of the intermediate tapered plate part **313** and an inner circumferential side of the outer tapered plate part **315**. A cross section of the outer curved plate part **314** in a plane including the central axis of the disc spring **116** has an arcuate shape that is convex toward the flexible disc **100** side. The outer curved plate part **314** has a circular shape centered on the central axis of the disc spring **116**. The outer curved plate part **314** is disposed on the same plane perpendicular to the center axis of the disc spring **116** over the entire circumference. The disc spring **116** is in contact with the flexible disc **100** over the entire circumference at the outer curved plate part **314**. The outer circumferential edge portion of the outer tapered plate part **315**, that is, the outer circumferential edge portion of the disc spring **116**, is spaced apart from the bottom part **122** and the flexible disc **100** in the axial direction over the entire circumference.
(72) The cap member main body **95**, the discs **97** to **99**, and the flexible disc **100** are radially positioned with respect to the mounting shaft part **28** of the piston rod **25** to be fixed. On the other hand, a range of movement of the disc spring **116** in a radial direction of the piston rod **25** is determined when the disc spring **116** comes into contact with the cylindrical part **124** of the cap member main body **95**. In other words, in the radial direction of the piston rod **25**, the disc spring **116** is movable slightly in the radial direction within an inner range of the cylindrical part **124** of the cap member main body **95** with respect to the cap member main body **95**, the discs **97** to **99**, and the flexible disc **100** which do not move. However, even if the disc spring **116** moves to the maximum in the radial direction, the disc spring **116** maintains a state in which the inner curved plate part **312** is in contact with a position of the bottom part **122** on a radially outer side of all the passage holes **126** of the bottom part **122** over the entire circumference and maintains a state in

which the outer curved plate part **314** is in contact with the flexible disc **100** over the entire circumference. An outer circumferential end portion of the disc spring **116** comes into contact with the cylindrical part **124** of the cap member main body **95** that covers the valve seat member **109**.
(73) The disc spring **117** is provided between the flexible disc **100** and the spring contact disc **104**. The disc spring **117** is a common part having the same shape as the disc spring **116**. The disc spring **117** is directed in a direction opposite to the disc spring **116** in the axial direction. The discs **101** to **103** are disposed on an inner side of the disc spring **117** in the radial direction with a gap with respect to an inner circumferential edge portion of disc spring **117** in the radial direction.
(74) The inner tapered plate part **311** of the disc spring **117** extends outward in the radial direction from an inner circumferential edge portion thereof to be inclined such that it is positioned further toward spring contact disc **104** side in the axial direction toward the outside in the radial direction. The intermediate tapered plate part **313** of the disc spring **117** extends outward in the radial direction from an outer circumferential side of the inner tapered plate part **311** to be inclined such that it is positioned further toward the flexible disc **100** side in the axial direction toward the outside in the radial direction. The outer tapered plate part **315** of the coned disc spring **117** extends outward in the radial direction from an outer circumferential side of the intermediate tapered plate part **313** to be inclined such that it is positioned further toward the spring contact disc **104** side in the axial direction toward the outside in the radial direction.
(75) The inner curved plate part **312** of the disc spring **117** is in contact with the spring contact disc **104** at a position on a radially outer side of all the passage holes **301** of the spring contact disc **104** over the entire circumference. The outer curved plate part **314** of the disc spring **117** is in contact with the flexible disc **100** over the entire circumference. An outer circumferential edge portion of disc spring **117** is spaced apart from spring contact disc **104** and the flexible disc **100** in the axial direction over the entire circumference. A range of movement of the disc spring **117** in the radial direction of the piston rod **25** is also determined when the disc spring **117** comes into contact with the cylindrical part **124** of the cap member main body **95**. Even if the disc spring **117** moves to the maximum within an inner range of the cylindrical part **124** of the cap member main body **95**, the disc spring **117** maintains a state in which the inner curved plate part **312** is in contact with the spring contact disc **104** at a position on a radially outer side of all the passage holes **301** of the spring contact disc **104** over the entire circumference and maintains a state in which the outer curved plate part **314** is in contact with the flexible disc **100** over the entire circumference. An outer circumferential end portion of the disc spring **117** comes into contact with the cylindrical part **124** of the cap member main body **95** that covers the valve seat member **109**.
(76) The spring member **105** includes a base plate part **331** having a bored disc shape fitted to the mounting shaft part **28**, and a plurality of spring plate parts **332** extending outward in a radial direction of the base plate part **331** from equally spaced positions of the base plate part **331** in the circumferential direction. The base plate part **331** has an outer diameter slightly smaller than twice a minimum distance from a center of the spring contact disc **104** to the passage hole **301**. The spring plate parts **332** are each inclined with respect to the base plate part **331** to become further away from the base plate part **331** in an axial direction of the base plate part **331** toward an extended distal end side. The spring member **105** is attached to the mounting shaft part **28** so that the spring plate part **332** extends from the base plate part **331** to the sub-valve **107** side in the axial direction of the base plate part **331**.
(77) The disc **106** has an outer diameter smaller than an outer diameter of the base plate part **331** of the spring member **105**. In the spring member **105**, the base plate part **331** is in contact with the disc **106**, and the plurality of spring plate parts **332** are in contact with the sub-valve **107**.
(78) As illustrated in FIG. **2**, the valve seat member **109** has a bored disc shape in which a through hole **131** extending in an axial direction of the valve seat member **109** and penetrating in a thickness direction to insert the mounting shaft part **28** therein is formed at a center in the radial direction. The through hole **131** includes a small diameter hole portion **132** on one side in the axial

direction into which the mounting shaft part **28** of the piston rod **25** is fitted, and a large diameter hole portion **133** on the other side in the axial direction having a larger diameter than the small diameter hole portion **132**.

(79) The valve seat member **109** includes an inner seat part **134** that has an annular shape to surround the large diameter hole portion **133** at an end portion on the large diameter hole portion **133** side in the axial direction. The valve seat member **109** includes a valve seat part **135** extending outward in the radial direction from the inner seat part **134**. The valve seat member **109** includes an inner seat part **138** that has an annular shape to surround the small diameter hole portion **132** at an end portion on the small diameter hole portion **132** side on a side opposite to the large diameter hole portion **133** in the axial direction. The valve seat member **109** includes a valve seat part **139** extending outward in the radial direction from the inner seat part **138**. A main body part **140** having a bored disc shape is provided between the inner seat part **134** and valve seat part **135**, and the inner seat part **138** and valve seat part **139** in the axial direction of the valve seat member **109**.

(80) The inner seat part **134** protrudes to one side in the axial direction of the main body part **140** from an inner circumferential edge portion of the main body part **140** on the large diameter hole portion **133** side in the axial direction. The valve seat part **135** also protrudes from the main body part **140** to the same side as the inner seat part **134** in the axial direction of the main body part **140** at an outer side of the inner seat part **134** in the radial direction. Distal end surfaces of the inner seat part **134** and the valve seat part **135** on the protruding side, that is, distal end surfaces thereof on a side opposite to the main body part **140**, are flat surfaces. The inner seat part **134** and the valve seat part **135** extend in a direction perpendicular to an axis of the valve seat member **109** and are disposed on the same plane.

(81) The inner seat part **138** protrudes to a side opposite to the inner seat part **134** in the axial direction of the main body part **140** from an inner circumferential edge portion of the main body part **140** on the small diameter hole portion **132** side in the axial direction. The valve seat part **139** also protrudes from the main body part **140** to the same side as the inner seat part **138** in the axial direction of the main body part **140** at an outer side of the inner seat part **138** in the radial direction. Distal end surfaces of the inner seat part **138** and the valve seat part **139** on the protruding side, that is, distal end surfaces thereof on a side opposite to the main body part **140**, are flat surfaces. The inner seat part **138** and the valve seat part **139** extend in a direction perpendicular to the axis of the valve seat member **109** and are disposed on the same plane. The inner seat parts **134** and **138** have the same outer diameter.

(82) The valve seat part **135** is a deformed seat having a petal shape. The valve seat part **135** includes a plurality of valve seat constituent parts **201** (only one is illustrated in FIG. **2** because it is a cross section). These valve seat constituent parts **201** have the same shape and are disposed at regular intervals in a circumferential direction of the valve seat member **109**. The inner seat part **134** has an annular shape centered on a central axis of the valve seat member **109**.

(83) Inside each of the valve seat constituent parts **201**, a passage recess **205** that is surrounded by the valve seat constituent part **201** and a part of the inner seat part **134** and is recessed in the axial direction of the valve seat member **109** from the distal end surfaces thereof on the protruding side is formed. A bottom surface of the passage recess **205** is formed with the main body part **140**. The passage recess **205** is formed on an inner side of all the valve seat constituent parts **201**.

(84) A passage hole **206** penetrating the valve seat member **109** in the axial direction by penetrating the main body part **140** in the axial direction is formed at a central position of the passage recess **205** in the circumferential direction of the valve seat member **109**.

(85) The passage hole **206** is a linear hole parallel to the central axis of the valve seat member **109**. The passage hole **206** is formed in the bottom surfaces of all the passage recesses **205**.

(86) The valve seat part **139** is also a deformed seat having a petal shape. The valve seat part **139** includes a plurality of valve seat constituent parts **211** (only one is illustrated in FIG. **2** because it is a cross section). These valve seat constituent parts **211** have the same shape and are disposed at

regular intervals in the circumferential direction of the valve seat member **109**. The valve seat constituent parts **211** have the same shape as the valve seat constituent parts **201**. The inner seat part **138** has an annular shape centered on the central axis of the valve seat member **109**.
(87) Inside each of the valve seat constituent parts **211**, a passage recess **215** that is surrounded by the valve seat constituent part **211** and a part of the inner seat part **138** and is recessed in the axial direction of the valve seat member **109** from the distal end surfaces thereof on the protruding side is formed. A bottom surface of the passage recess **215** is formed with the main body part **140**. The passage recess **215** is formed on an inner side of all the valve seat constituent parts **211**.
(88) A passage hole **216** penetrating the valve seat member **109** in the axial direction by penetrating the main body part **140** in the axial direction is formed at a central position of the passage recess **215** in the circumferential direction of the valve seat member **109**.
(89) The passage hole **216** is a linear hole parallel to the central axis of the valve seat member **109**. The passage hole **216** is formed in the bottom surfaces of all the passage recesses **215**.
(90) A disposition pitch of the plurality of valve seat constituent parts **201** in the circumferential direction of the valve seat member **109** is the same as a disposition pitch of the plurality of valve seat constituent parts **211** in the circumferential direction of the valve seat member **109**. The valve seat constituent parts **201** and the valve seat constituent parts **211** are shifted from each other by half a pitch. The passage hole **206** is disposed between the valve seat constituent parts **211** adjacent to each other in the circumferential direction of the valve seat member **109**. Therefore, the passage hole **206** is disposed outside a range of the valve seat part **139**. The passage hole **216** is disposed between the valve seat constituent parts **201** adjacent to each other in the circumferential direction of the valve seat member **109**. Therefore, the passage hole **216** is disposed outside a range of the valve seat part **135**.
(91) A passage groove **221** crossing the inner seat part **134** in the radial direction is formed in the valve seat member **109** on the large diameter hole portion **133** side in the axial direction. The passage groove **221** is formed to be recessed in the axial direction of the valve seat member **109** from the distal end surface of the inner seat part **134** on a side opposite to the main body part **140**. The passage groove **221** also includes a space between the valve seat constituent parts **201** adjacent to each other in the circumferential direction of the valve seat member **109**. The passage hole **216** opens to the bottom surface of the passage groove **221**. The passage groove **221** allows the passage hole **216** and the large diameter hole portion **133** to communicate with each other.
(92) The passage hole **216** and the passage recess **215** to which the passage hole **216** opens form a first passage part **161** provided in the valve seat member **109**. A plurality of first passage parts **161** are provided in the valve seat member **109** at regular intervals in the circumferential direction of the valve seat member **109**. The passage groove **221** forms a radial passage **222** extending in the radial direction toward the first passage part **161**. A plurality of radial passages **222** are provided in the valve seat member **109** at regular intervals in the circumferential direction of the valve seat member **109**.
(93) The valve seat member **109** includes a passage groove **225** formed between valve seat constituent parts **211** adjacent to each other in the circumferential direction of the valve seat member **109**. The passage hole **206** opens to a bottom surface of the passage groove **225**. Therefore, the passage groove **225** communicates with the passage hole **206**.
(94) The passage hole **206** and the passage recess **205** to which the passage hole **206** opens form a second passage part **162** provided in the valve seat member **109**. A plurality of second passage parts **162** are provided in the valve seat member **109** at regular intervals in the circumferential direction of the valve seat member **109**.
(95) The plurality of first passage parts **161** and the plurality of second passage parts **162** constitute a valve seat member passage part **160** provided in the valve seat member **109** to allow the oil fluid L to flow.
(96) In the valve seat member **109**, an annular seal groove **141** recessed inward in the radial

direction is formed at an axially intermediate position of an outer circumferential portion of the main body part **140**. The O-ring **108** is disposed in the seal groove **141**. The valve seat member **109** is fitted to the cylindrical part **124** of the cap member main body **95** at an outer circumferential portion thereof with the inner seat part **138** and the valve seat part **139** directed toward a side opposite to the bottom part **122**. In this state, the O-ring **108** seals a gap between the cylindrical part **124** of the cap member main body **95** and the valve seat member **109**.
(97) The cap member main body **95**, the O-ring **108**, and the valve seat member **109** form a cap chamber **146** on an inner side of the cap member main body **95**. The cap chamber **146** is provided between the bottom part **122** and the valve seat member **109** of the cap member main body **95**. As illustrated in FIG. **3**, the discs **97** to **99**, **101** to **103**, and **106**, the flexible disc **100**, the spring contact disc **104**, the spring member **105**, the sub-valve **107**, and the disc springs **116** and **117** are provided inside the cap chamber **146**.
(98) A lower chamber communication chamber **149** is formed in the cap chamber **146** by being surrounded by the flexible disc **100**, the disc spring **116**, the discs **97** to **99**, and the bottom part **122** of the cap member main body **95**. The lower chamber communication chamber **149** is in constant communication with communication passages **148** in the plurality of passage holes **126** of the bottom part **122** of the cap member main body **95**.
(99) An upper chamber communication chamber **147** is formed in the cap chamber **146** by being surrounded by the flexible disc **100**, the disc spring **117**, the discs **101** to **103**, and the spring contact disc **104**. The upper chamber communication chamber **147** is in constant communication with the communication passages **302** in the plurality of passage holes **301** of the spring contact disc **104**. Communication between the lower chamber communication chamber **149** and the upper chamber communication chamber **147** is blocked by the flexible disc **100**.
(100) As illustrated in FIG. **2**, the annular valve seat member **109** and the bottomed cylindrical cap member **305** are disposed in the lower chamber **23**. At this time, in the valve seat member **109**, the valve seat part **135** is disposed on the cap chamber **146** side and the valve seat part **139** is disposed on the lower chamber **23** side. The communication passages **148** of the bottom part **122** of the cap member main body **95** are in constant communication with the lower chamber **23**.
(101) A space between the valve seat member **109** of the cap chamber **146** and the spring contact disc **104** is an intermediate chamber **150** that is in constant communication with the upper chamber communication chamber **147** via the communication passages **302**. The intermediate chamber **150** is in constant communication with the upper chamber **22** through the radial passage **222** in the passage groove **221** of the valve seat member **109**, the passage in the large diameter hole portion **133** of the valve seat member **109**, the piston rod passage part **51** in the passage notch part **30** of the piston rod **25** and the passage in the large diameter hole portion **46** of the piston **21**, the passage in the notch part **90** of the disc **82**, and the passages in the annular groove **55** and the plurality of passage holes **38** of the piston **21**. Therefore, the upper chamber communication chamber **147** is in constant communication with the upper chamber **22** via the intermediate chamber **150**.
(102) Volumes of the lower chamber communication chamber **149** and the upper chamber communication chamber **147** change due to the flexible disc **100** bending in the axial direction. That is, bending of the flexible disc **100** causes the lower chamber communication chamber **149** and the upper chamber communication chamber **147** to have a function of an accumulator. The lower chamber communication chamber **149** decreases in volume and discharges the oil fluid L to the lower chamber **23** to absorb an increase in volume of the upper chamber communication chamber **147**, or increases in volume and allows the oil fluid L to flow in from the lower chamber **23** to absorb a decrease in volume of the upper chamber communication chamber **147**. Conversely, the upper chamber communication chamber **147** decreases in volume and discharges the oil fluid L to the upper chamber **22** side to absorb an increase in volume of the lower chamber communication chamber **149**, or increases in volume and allows the oil fluid L to flow in from the upper chamber **22** to absorb a decrease in volume of the lower chamber communication chamber **149**.

(103) As described above, deformation of the flexible disc **100** being hindered by the oil fluid L in the upper chamber communication chamber **147** and the lower chamber communication chamber **149** is suppressed.
(104) The valve seat member **109** partitions the intermediate chamber **150** and the lower chamber **23**. The valve seat member **109** is provided to face both the intermediate chamber **150** and the lower chamber **23**. A plurality of passage grooves **225** are provided to face the lower chamber **23**. The plurality of second passage parts **162** are in constant communication with the lower chamber **23** via passages in the plurality of passage grooves **225**. The communication passages **148** formed in the bottom part **122** of the cap member main body **95** are in constant communication with the lower chamber **23** which is one of the upper chamber **22** and the lower chamber **23**.
(105) The radial passage **222** in the passage groove **221** that opens to the first passage parts **161** of the valve seat member **109** is in constant communication with the upper chamber communication chamber **147** via the intermediate chamber **150** and the communication passages **302** of the spring contact disc **104**. The radial passage **222** allows the inside of the upper chamber communication chamber **147** to constantly communicate with the passage in the large diameter hole portion **133** of the valve seat member **109** and the piston rod passage part **51** in the passage notch part **30** of the piston rod **25**.
(106) The sub-valve **107** has a disc shape and has an outer diameter equal to an outer diameter of the valve seat part **135** of the valve seat member **109**. The sub-valve **107** is in constant contact with the inner seat part **134** and can be separated from and seated on the valve seat part **135**. The sub-valve **107** closes all the second passage parts **162** by being seated on the entire valve seat part **135**. The sub-valve **107** is seated entirely on one of the valve seat constituent parts **201** of the valve seat part **135** to close the second passage parts **162** inside the valve seat constituent part **201**. The spring member **105** brings the sub-valve **107** into contact with the valve seat part **135** of the valve seat member **109**. The sub-valve **107** is seated on the valve seat part **135** by a biasing force of the spring member **105** to close the second passage parts **162**.
(107) The sub-valve **107** that can be separated from and seated on the valve seat part **135** is provided in the cap chamber **146**. The sub-valve **107** allows communication between the plurality of second passage parts **162**, the intermediate chamber **150**, and the upper chamber communication chamber **147** by being separated from the valve seat part **135** in the cap chamber **146**. As a result, the lower chamber **23** communicates with the upper chamber **22**. At this time, the sub-valve **107** generates a damping force by suppressing a flow of the oil fluid L between itself and the valve seat part **135**. The sub-valve **107** is an inflow valve that opens when the oil fluid L is caused to flow from the lower chamber **23** to a side of the intermediate chamber **150** and the upper chamber communication chamber **147** through the plurality of second passage parts **162**. The sub-valve **107** is a check valve that restricts an outflow of the oil fluid L from the intermediate chamber **150** and the upper chamber communication chamber **147** to the lower chamber **23** via the second passage parts **162**. The passage hole **216** constituting the first passage part **161** opens outside the range of the valve seat part **135** in the valve seat member **109**. Therefore, the passage hole **216** is in constant communication with the intermediate chamber **150** and the upper chamber communication chamber **147** regardless of the sub-valve **107** seated on the valve seat part **135**.
(108) The passages in the plurality of passage grooves **225**, the plurality of second passage parts **162**, the passage between the sub-valve **107** and the valve seat part **135** that appears when the valve opens, the intermediate chamber **150**, the upper chamber communication chamber **147**, the radial passage **222** in the passage groove **221** of the valve seat member **109**, the passage in the large diameter hole portion **133** of the valve seat member **109**, the piston rod passage part **51** in the passage notch part **30** of the piston rod **25** and the passage in the large diameter hole portion **46** of the piston **21**, the passage in the notch part **90** of the disc **82**, and the passages in the annular groove **55** and the plurality of passage holes **38** of the piston **21** constitute a second passage **172** through which the oil fluid L flows from the lower chamber **23** on the upstream side to the upper chamber

**22** on the downstream side in the cylinder **4** due to the piston **21** moving to the lower chamber **23** side. The second passage **172** serves as a compression-side passage through which the oil fluid L flows from the lower chamber **23** on the upstream side toward the upper chamber **22** on the downstream side when the piston **21** moves to the lower chamber **23** side, that is, during the compression stroke. The intermediate chamber **150** and the upper chamber communication chamber **147** are provided in the second passage **172**.

(109) The spring contact disc **104** is thicker and more rigid than the sub-valve **107**. When the sub-valve **107** is deformed, the spring contact disc **104** comes into contact with the sub-valve **107** to restrict further deformation of the sub-valve **107**. The sub-valve **107**, the valve seat member **109** including the valve seat part **135**, the cap member main body **95**, the disc **106**, the spring member **105**, and the spring contact disc **104** constitute a second damping force generation mechanism **173**. The second damping force generation mechanism **173** is provided in the compression-side second passage **172**, and opens and closes the second passage **172** to suppress a flow of the oil fluid L from the second passage **172** to the upper chamber **22**, thereby generating a damping force.

(110) The second damping force generation mechanism **173** is provided in the piston rod **25**. The valve seat part **135** of the second damping force generation mechanism **173** is provided in the valve seat member **109**. The second damping force generation mechanism **173** is disposed separately from the first damping force generation mechanism **42** that generates a damping force in the same compression stroke. The sub-valve **107** constituting the compression-side second damping force generation mechanism **173** serves as a compression-side sub-valve. The cap member **305** covers one end side of the second damping force generation mechanism **173** and an outer circumferential side of the valve seat member **109**. The cap member **305** may cover one end side of the second damping force generation mechanism **173** and at least a part of the outer circumference of the valve seat member **109**.

(111) The communication passages **148** that allow the inside and outside of the cap member **305** to communicate are formed in the bottom part **122** on one end side of the cap member **305** in the axial direction. The disc spring **116** is provided so that one end surface side in the axial direction (thickness direction) is in contact with an outer circumferential side of the cap member **305** with respect to the communication passages **148**. The bendable flexible disc **100** is provided to be in contact with the other end surface side of the disc spring **116** in the axial direction (thickness direction). The disc spring **116** has a plate shape, and is formed so that an outer circumferential surface side is bent to be in contact with the flexible disc **100** and an inner circumferential surface side is bent to be in contact with the bottom part **122** of the cap member **305**.

(112) The communication passages **302** that allow the inside and outside of the cap member **305** to communicate are formed in the spring contact disc **104** that is a bottom part of the cap member **305** on the other end side in the axial direction. The disc spring **117** is provided so that one end surface side in the axial direction (thickness direction) is in contact with the outer circumferential side of the cap member **305** with respect to the communication passages **302**. The bendable flexible disc **100** is provided to be in contact with the other end surface side of the disc spring **117** in the axial direction (thickness direction). That is, the disc spring **116** and the disc spring **117** are provided on both one end surface side and the other end surface side of the flexible disc **100**. The disc spring **117** has a plate shape. The disc spring **117** is formed so that an outer circumferential surface side is bent to be in contact with the flexible disc **100** and an inner circumferential surface side is bent to be in contact with the spring contact disc **104** of the cap member **305**.

(113) In the second passage **172**, when the second damping force generation mechanism **173** is in an open state, the passage in the notch part **90** of the disc **82** is the narrowest in portions having a fixed flow path cross-sectional area and has a smaller flow path cross-sectional area than an upstream side and a downstream side thereof, thereby serving as an orifice **175** in the second passage **172**. The orifice **175** is disposed downstream of the sub-valve **107** in the flow of the oil fluid L when the sub-valve **107** opens and the oil fluid L flows through the second passage **172**.

The orifice **175** may be disposed upstream of the sub-valve **107** in the flow of the oil fluid L when the sub-valve **107** opens and the oil fluid L flows through the second passage **172**. The orifice **175** is formed by cutting out the disc **82** of the first damping force generation mechanism **41** in contact with the piston **21**.

(114) In the compression-side second damping force generation mechanism **173**, a fixed orifice that allows communication between the upper chamber **22** and the lower chamber **23** even in a state in which the valve seat part **135** and the sub-valve **107** that is in contact with the valve seat part **135** are in contact with each other is not formed in either of the valve seat part **135** and the sub-valve **107**. That is, the compression-side second damping force generation mechanism **173** does not allow the upper chamber **22** and the lower chamber **23** to communicate with each other in a state in which the valve seat part **135** and the sub-valve **107** are in contact with each other over the entire circumference. In other words, the second passage **172** does not have a fixed orifice formed to allow constant communication between the upper chamber **22** and the lower chamber **23** and is not a passage that allows constant communication between the upper chamber **22** and the lower chamber **23**.

(115) The compression-side second passage **172** that allows communication between the upper chamber **22** and the lower chamber **23** is in parallel with the first passage **72** serving as a compression-side passage that similarly allows communication between the upper chamber **22** and the lower chamber **23**, the first damping force generation mechanism **42** is provided in the first passage **72**, and the second damping force generation mechanism **173** is provided in the second passage **172**. Therefore, the first damping force generation mechanism **42** and the second damping force generation mechanism **173**, which are both on the compression side, are disposed in parallel.

(116) As illustrated in FIG. **3**, the sub-valve **110** has a disc shape. The sub-valve **110** has the same outer diameter as the valve seat part **139** of the valve seat member **109**. The sub-valve **110** is in constant contact with the inner seat part **138** and can be separated from and seated on the valve seat part **139**. The sub-valve **110** closes all the first passage parts **161** by being seated on the entire valve seat part **139**. The sub-valve **110** is seated entirely on one of the valve seat constituent parts **211** of the valve seat part **139** to close the first passage part **161** inside the valve seat constituent part **211**. The sub-valve **110** can be a common part having the same shape as the sub-valve **107**.

(117) The disc **111** is a common part having the same shape as the disc **106**. The disc **111** has an outer diameter smaller than the outer diameter of the sub-valve **110** and smaller than an outer diameter of the inner seat part **138**.

(118) The spring member **112** includes the base plate part **341** having a bored disc shape fitted to the mounting shaft part **28**, and a plurality of spring plate parts **342** extending outward in a radial direction of the base plate part **341** from equally spaced positions of the base plate part **341** in the circumferential direction. The base plate part **341** has an outer diameter larger than the outer diameter of the disc **111**. The spring plate parts **342** are each inclined with respect to the base plate part **341** to become further away from the base plate part **341** in an axial direction of the base plate part **341** toward an extended distal end side. The spring member **112** is attached to the mounting shaft part **28** so that the spring plate parts **342** extend from the base plate part **341** to the sub-valve **110** side in the axial direction of the base plate part **341**. In the spring member **112**, the base plate part **341** is in contact with the disc **111**, and the plurality of spring plate parts **342** are in contact with the sub-valve **110**. The spring member **112** brings the sub-valve **110** into contact with the valve seat part **139** of the valve seat member **109**. The sub-valve **110** is seated on the valve seat part **139** by a biasing force of the spring member **112** to close the first passage parts **161**.

(119) The sub-valve **110** is provided in the lower chamber **23**. The sub-valve **110** allows the upper chamber **22**, the intermediate chamber **150**, and the upper chamber communication chamber **147** to communicate with the lower chamber **23** by being separated from the valve seat part **139**. At this time, the sub-valve **110** generates a damping force by suppressing a flow of the oil fluid L between itself and the valve seat part **139**. The sub-valve **110** is a discharge valve that opens when the oil

fluid L is discharged from the inside of the upper chamber **22**, the intermediate chamber **150**, and the upper chamber communication chamber **147** to the lower chamber **23** through the plurality of first passages **161** of the valve seat member **109**. The sub-valve **110** is a check valve that restricts an inflow of the oil fluid L via the first passage parts **161** from the lower chamber **23** into the upper chamber **22**, the intermediate chamber **150**, and the upper chamber communication chamber **147**. As illustrated in FIG. **2**, the passage hole **206** constituting the second passage part **162** opens outside the range of the valve seat part **139** in the valve seat member **109**. Therefore, the passage hole **206** is in constant communication with the lower chamber **23** regardless of the sub-valve **110** seated on the valve seat part **139**.
(120) The passages in the plurality of passage holes **38** and the annular groove **55** of the piston **21**, the passage in the notch part **90** of the disc **82**, the piston rod passage part **51** in the passage notch part **30** of the piston rod **25**, the passage in the large diameter hole portion **46** of the piston **21** and the passage in the large diameter hole portion **133** of the valve seat member **109**, the radial passage **222** in the passage groove **221** of the valve seat member **109**, the intermediate chamber **150**, the upper chamber communication chamber **147**, the plurality of first passages **161** of the valve seat member **109**, and the passage between the sub-valve **110** and the valve seat part **139** that appears when the valve opens constitute a second passage **182** through which the oil fluid L flows from the upper chamber **22** on the upstream side to the lower chamber **23** on the downstream side in the cylinder **4** due to the piston **21** moving to the upper chamber **22** side. The second passage **182** serves as an extension-side passage through which the oil fluid L flows from the upper chamber **22** on the upstream side toward the lower chamber **23** on the downstream side when the piston **21** moves to the upper chamber **22** side, that is, during the extension stroke.
(121) The disc **113** has an outer diameter equal to the outer diameter of the sub-valve **110**. The disc **113** is thicker and more rigid than the sub-valve **110**. When the sub-valve **110** is deformed, the disc **113** comes into contact with the sub-valve **110** to restrict further deformation of the sub-valve **110**. The annular member **114** has an outer diameter smaller than the outer diameter of the disc **113**. The annular member **114** is a common part having the same shape as the annular member **69**.
(122) The sub-valve **110**, the valve seat member **109** including the valve seat part **139**, the discs **111** and **113**, and the spring member **112** constitute an extension-side second damping force generation mechanism **183**. The second damping force generation mechanism **183** is provided in the extension-side second passage **182**, and opens and closes the second passage **182** to suppress a flow of the oil fluid L from the second passage **182** to the lower chamber **23**, thereby generating a damping force. In other words, the second damping force generation mechanism **183** is provided in the piston rod **25**, and the valve seat part **139** thereof is provided in the valve seat member **109**. The second damping force generation mechanism **183** is disposed separately from the first damping force generation mechanism **41** that generates a damping force in the same extension stroke. The sub-valve **110** constituting the extension-side second damping force generation mechanism **183** serves as an extension-side sub-valve.
(123) The lower chamber communication chamber **149** communicating with the lower chamber **23** via the communication passages **148** is disposed in parallel with the second passage **172** and the second passage **182**. The flexible disc **100**, the disc springs **116** and **117**, the discs **97** to **99**, and the lower chamber communication chamber **149** constitute a lower chamber volume variable mechanism **185** capable of changing a volume of the lower chamber communication chamber **149**. In terms of the flow path, the lower chamber communication chamber **149** is provided between the flexible disc **100** and the sub-valve **110** via the lower chamber **23** and the communication passages **148**. The flexible disc **100** changes a volume of the lower chamber communication chamber **149** to be increased by being deformed and moved away from the bottom part **122**, and changes the volume of the lower chamber communication chamber **149** to be reduced by being deformed and moved closer to the bottom part **122**.
(124) The intermediate chamber **150** and the upper chamber communication chamber **147** which

communicate with the upper chamber **22** constitute the second passage **172** and the second passage **182**. The flexible disc **100**, the disc springs **116** and **117**, the discs **101** to **103**, and the upper chamber communication chamber **147** constitute an upper chamber volume variable mechanism **186** capable of changing a volume of the upper chamber communication chamber **147**. In terms of flow path, the upper chamber communication chamber **147** is provided between the flexible disc **100** and the sub-valve **107**. The upper chamber volume variable mechanism **186** changes a volume of the upper chamber communication chamber **147** to be increased by deforming and moving the flexible disc **100** away from the spring contact disc **104**, and changes the volume of the upper chamber communication chamber **147** to be reduced by deforming and moving the flexible disc **100** closer to the spring contact disc **104**.

(125) The flexible disc **100** and the disc springs **116** and **117** are commonly used for the lower chamber volume variable mechanism **185** and the upper chamber volume variable mechanism **186**. The lower chamber volume variable mechanism **185** including the lower chamber communication chamber **149** and the upper chamber volume variable mechanism **186** including the upper chamber communication chamber **147** constitute an accumulator **190**.

(126) In the second passage **182**, when the second damping force generation mechanism **183** is in an open state, the passage in the notch part **90** of the disc **82** is the narrowest in portions having a fixed flow path cross-sectional area and has a smaller flow path cross-sectional area than an upstream side and a downstream side thereof, thereby serving as the orifice **175** also in the second passage **182**. The orifice **175** is common to the second passages **172** and **182**. The orifice **175** is disposed upstream of the sub-valve **110** in the flow of the oil fluid L when the sub-valve **110** opens and the oil fluid L flows through the second passage **182**. The orifice **175** may be disposed downstream of the sub-valve **110** in the flow of the oil fluid L when the sub-valve **110** opens and the oil fluid L flows through the second passage **182**. The sub-valve **110** and the above-described sub-valve **107** open and close independently.

(127) In the extension-side second damping force generation mechanism **183**, a fixed orifice that allows communication between the upper chamber **22** and the lower chamber **23** even in a state in which the valve seat part **139** and the sub-valve **110** that is in contact with the valve seat part **139** are in contact with each other is not formed in either of the valve seat part **139** and the sub-valve **110**. That is, the extension-side second damping force generation mechanism **183** does not allow the upper chamber **22** and the lower chamber **23** to communicate with each other if the valve seat part **139** and the sub-valve **110** are in contact with each other over the entire circumference. In other words, the second passage **182** does not have a fixed orifice formed to allow constant communication between the upper chamber **22** and the lower chamber **23** and is not a passage that allows constant communication between the upper chamber **22** and the lower chamber **23**. The annular member **114**, together with the disc **113**, restricts deformation of the sub-valve **110** beyond a specified limit in an opening direction by coming into contact with the sub-valve **110**.

(128) In the shock absorber **1**, as a flow of allowing the oil fluid L to pass in the axial direction at least within a range of the piston **21**, the upper chamber **22** and the lower chamber **23** can communicate only through the first damping force generation mechanisms **41** and **42** and the second damping force generation mechanisms **173** and **183**. In the shock absorber **1**, a fixed orifice that allows constant communication between the upper chamber **22** and the lower chamber **23** is not provided on the passage of the oil fluid L.

(129) The extension-side second passage **182** that allows communication between the upper chamber **22** and the lower chamber **23** is in parallel with the first passage **92** serving as an extension-side passage that similarly allows communication between the upper chamber **22** and the lower chamber **23** except for the passages in the annular groove **55** and the plurality of passage holes **38** on the upper chamber **22** side. In the portion in parallel, the first damping force generation mechanism **41** is provided in the first passage **92**, and the second damping force generation mechanism **183** is provided in the second passage **182**. Therefore, the first damping force

generation mechanism **41** and the second damping force generation mechanism **183**, which are both on the extension side, are disposed in parallel.
(130) The second damping force generation mechanisms **173** and **183** include the valve seat member **109**, the sub-valve **110** provided on one side of the valve seat member passage part **160** and the sub-valve **107** provided on the other side of the valve seat member passage part **160** which is a portion of the second passages **172** and **182** provided in the valve seat member **109**, and the bottomed cylindrical cap member **305** provided between the piston **21** and the valve seat member **109** in the second passages **172** and **182**. The valve seat member **109** is provided in the cap member **305**. The sub-valve **110** is provided on the lower chamber **23** side of the valve seat member **109**. The sub-valve **107** is provided in the cap chamber **146** between the bottom part **122** of the cap member **305** and the valve seat member **109**.
(131) In a state in which the main valve **71** is assembled to the piston rod **25**, the main valve **71** is clamped by the disc **63** and the disc **67** on the inner circumferential side, and is in contact with the valve seat part **50** of the piston **21** on the outer circumferential side over the entire circumference. Also, in this state, the main valve **91** is clamped by the disc **83** and the disc **87** on the inner circumferential side, and is in contact with the valve seat part **48** of the piston **21** on the outer circumferential side over the entire circumference.
(132) In this state, the sub-valve **107** is clamped by the inner seat part **134** of the valve seat member **109** and the disc **106** on the inner circumferential side and is in contact with the valve seat part **135** of the valve seat member **109** over the entire circumference. Also, in this state, the sub-valve **110** is clamped by the inner seat part **138** of the valve seat member **109** and the disc **111** on the inner circumferential side and is in contact with the valve seat part **139** of the valve seat member **109** over the entire circumference.
(133) In this state, as illustrated in FIG. **3**, the flexible disc **100** is clamped by the discs **99** and **101** on the inner circumferential side, and is in contact with the outer curved plate part **314** of the disc spring **116** and the outer curved plate part **314** of the disc spring **117** on the outer circumferential side over the entire circumference. Also, in this state, the inner curved plate part **312** of the disc spring **116** is in contact with the bottom part **122** of the cap member main body **95** over the entire circumference while the disc spring **116** is elastically deformed, and the inner curved plate part **312** of the disc spring **117** is in contact with the spring contact disc **104** over the entire circumference while the disc spring **117** is elastically deformed.
(134) As illustrated in FIG. **1**, a fluid passage **251** and a fluid passage **252** are formed to penetrate the valve body **12** in the axial direction. The fluid passages **251** and **252** allow communication between the lower chamber **23** and the reservoir chamber **5**. The base valve **15** includes a valve mechanism **255** capable of opening and closing the fluid passage **251** with respect to the bottom member **9** side of the valve body **12** in the axial direction. The base valve **15** includes a valve mechanism **256** capable of opening and closing the fluid passage **252** with respect to a side of the valve body **12** opposite to the bottom member **9** in the axial direction.
(135) In the base valve **15**, when the piston rod **25** moves to the compression side to move the piston **21** in a direction that reduces the lower chamber **23** and a pressure in the lower chamber **23** becomes higher than a pressure in the reservoir chamber **5** by a predetermined value or higher, the valve mechanism **255** opens the fluid passage **251** to allow the oil fluid L in the lower chamber **23** to flow into the reservoir chamber **5** and generates a damping force at that time. In other words, when the piston rod **25** moves to the compression side to move the piston **21**, the oil fluid L flows into the reservoir chamber **5** in the fluid passage **251**. The valve mechanism **255** is a compression-side damping force generation mechanism. This valve mechanism **255** does not hinder a flow of the oil fluid L in the fluid passage **252**.
(136) In the base valve **15**, when the piston rod **25** moves to the extension side to move the piston **21** to the upper chamber **22** side and a pressure in the lower chamber **23** becomes lower than a pressure in the reservoir chamber **5**, the valve mechanism **256** opens the fluid passage **252** to allow

the oil fluid L in the reservoir chamber **5** to flow into the lower chamber **23**. In other words, when the piston rod **25** moves to the extension side to move the piston **21**, the oil fluid L flows into the lower chamber **23** in the fluid passage **252**. The valve mechanism **256** is a suction valve that allows the oil fluid L to flow from the reservoir chamber **5** into the lower chamber **23** without substantially generating a damping force at that time. This valve mechanism **256** does not hinder a flow of the oil fluid L in the fluid passage **251**.

(137) <Operation>

(138) As illustrated in FIG. **2**, the main valve **91** of the first damping force generation mechanism **41** has higher rigidity and a higher valve opening pressure than the sub-valve **110** of the second damping force generation mechanism **183**. Therefore, in the extension stroke, in a very low speed region in which the piston speed is lower than a predetermined value, the second damping force generation mechanism **183** opens while the first damping force generation mechanism **41** is in a closed state. Also, in a normal speed region in which the piston speed is equal to or higher than the predetermined value, both the first damping force generation mechanism **41** and the second damping force generation mechanism **183** open. The sub-valve **110** is a very low speed valve that opens in a region in which the piston speed is very low to generate a damping force.

(139) That is, in the extension stroke, a pressure of the upper chamber **22** increases and a pressure of the lower chamber **23** decreases when the piston **21** moves to the upper chamber **22** side. Then, although none of the first damping force generation mechanisms **41** and **42** and the second damping force generation mechanisms **173** and **183** has a fixed orifice that allows constant communication between the upper chamber **22** and the lower chamber **23**, the oil fluid L in the upper chamber **22** flows to the intermediate chamber **150** via the passages in the plurality of passage holes **38** and the annular groove **55** of the piston **21**, the orifice **175**, the passage in the large diameter hole portion **46** of the piston **21**, the piston rod passage part **51** in the passage notch part **30** of the piston rod **25** and the passage in the large diameter hole portion **133** of the valve seat member **109**, and the radial passage **222** in the passage groove **221** of the valve seat member **109**, and then flows into the upper chamber communication chamber **147** via the communication passage **302** of the spring contact disc **104**. Thereby, the upper chamber communication chamber **147** rises in pressure. Therefore, in the upper chamber volume variable mechanism **186**, a portion of the flexible disc **100** on a radially inner side of a position in contact with the outer curved plate part **314** of the disc spring **116** bends to the bottom part **122** side to increase a volume of the upper chamber communication chamber **147**. Thereby, the upper chamber volume variable mechanism **186** suppresses a pressure rise in the upper chamber communication chamber **147**. At this time, since the flexible disc **100** bends and moves to the bottom part **122** side, the lower chamber volume variable mechanism **185** reduces a volume of the lower chamber communication chamber **149**.

(140) Here, in the extension stroke of the shock absorber **1** at the time of a low-frequency input (at the time of a large amplitude of vibration), since an amount of the oil fluid L flowing from the upper chamber **22** to the upper chamber communication chamber **147** as described above becomes large, the flexible disc **100** is greatly deformed. If an amount of deformation of the flexible disc **100** becomes large, a reaction force due to support stiffness of the clamped inner circumferential side increases, and the amount of deformation is restricted. Thereby, the intermediate chamber **150** and the upper chamber communication chamber **147** rise in pressure. As a result, the second passage **182** rises in pressure until reaching a state in which the second damping force generation mechanism **183** opens.

(141) At this time, since none of the first damping force generation mechanisms **41** and **42** and the second damping force generation mechanisms **173** and **183** has a fixed orifice that allows constant communication between the upper chamber **22** and the lower chamber **23**, in the extension stroke having a piston speed lower than a first predetermined value at which the second damping force generation mechanism **183** opens, the damping force rises sharply. Also, in a very low speed region in which the piston speed is in a region higher than the first predetermined value and is lower than

a second predetermined value that is a higher speed than the first predetermined value, the second damping force generation mechanism **183** opens while the first damping force generation mechanism **41** is in a closed state.
(142) That is, the sub-valve **110** is separated from the valve seat part **139** to allow the upper chamber **22** and the lower chamber **23** to communicate with each other through the extension-side second passage **182**. Therefore, the oil fluid L in the upper chamber **22** flows to the lower chamber **23** through the passages in the plurality of passage holes **38** and the annular groove **55** of the piston **21**, the orifice **175**, the passage in the large diameter hole portion **46** of the piston **21**, the piston rod passage part **51** in the passage notch part **30** of the piston rod **25** and the passage in the large diameter hole portion **133** of the valve seat member **109**, the radial passage **222** in the passage groove **221** of the valve seat member **109**, the intermediate chamber **150**, the first passage part **161** in the valve seat member **109**, and the passage between the sub-valve **110** and the valve seat part **139**.
(143) Thereby, even in the very low speed region in which the piston speed is lower than the second predetermined value, a damping force due to valve characteristics (characteristics in which a damping force is substantially proportional to a piston speed) can be obtained.
(144) In the extension stroke, in a normal speed region in which the piston speed is equal to or higher than the second predetermined value, the first damping force generation mechanism **41** opens while the second damping force generation mechanism **183** remains in an open state. That is, the sub-valve **110** is separated from the valve seat part **139** as described above to allow the oil fluid L to flow from the upper chamber **22** to the lower chamber **23** through the extension-side second passage **182**. At this time, when the flow of the oil fluid L is reduced by the orifice **175** provided downstream of the sub-valve **91** in the second passage **182**, a pressure applied to the main valve **91** becomes high to increase a differential pressure, and the main valve **91** is separated from the valve seat part **48** to allow the oil fluid L to flow from the upper chamber **22** to the lower chamber **23** through the extension-side first passage **92**. Therefore, the oil fluid L in the upper chamber **22** flows to the lower chamber **23** through the passages in the plurality of passage holes **38** and the annular groove **55**, and the passage between the main valve **91** and the valve seat part **48**.
(145) Thereby, even in the normal speed region in which the piston speed is equal to or higher than the second predetermined value, a damping force due to valve characteristics (a damping force is substantially proportional to a piston speed) can be obtained. A rate of increase in the extension-side damping force with respect to an increase in the piston speed in the normal speed region is lower than a rate of increase in the extension-side damping force with respect to an increase in the piston speed in the very low speed region. In other words, a slope of the rate of increase in the extension-side damping force with respect to the increase in piston speed in the normal speed region can be laid down more than that in the very low speed region.
(146) Here, in the extension stroke, in the normal speed region in which the piston speed is equal to or higher than the second predetermined value, a differential pressure between the upper chamber **22** and the lower chamber **23** is larger than that in the low speed region in which the piston speed is equal to or higher than the first predetermined value and lower than the second predetermined value, but since the first passage **92** is not narrowed by an orifice, the oil fluid L can be allowed to flow at a high flow rate through the first passage **92** when the main valve **91** opens. In addition to this, when the second passage **182** is narrowed by the orifice **175**, deformation of the sub-valve **110** can be suppressed.
(147) Also, at this time, the sub-valve **107** in a closed state receives a pressure in an opposite direction from the lower chamber **23** and the intermediate chamber **150**. Even if the differential pressure between the upper chamber **22** and the lower chamber **23** increases, since the orifice **175** is formed upstream of the sub-valve **107** in the second passage **182**, a pressure rise in the intermediate chamber **150** becomes gentle with respect to a pressure rise in the upper chamber **22**, and the pressure difference between the intermediate chamber **150** and the lower chamber **23** is

inhibited from increasing. Therefore, the pressure difference between the intermediate chamber **150** and the lower chamber **23** received by the sub-valve **107** in a closed state is inhibited from increasing, and a large back pressure from the intermediate chamber **150** side toward the lower chamber **23** side can be inhibited from being applied to the sub-valve **107**.
(148) In the shock absorber **1**, the flow path for allowing the oil fluid L to flow from the upper chamber **22** to the lower chamber **23** in the extension stroke is provided in parallel by the first passage **92** and the second passage **182**, and the main valve **91** and the sub-valve **110** are provided in parallel. Also, the orifice **175** is connected in series to the sub-valve **110**.
(149) As described above, in the extension stroke, in the normal speed region in which the piston speed is equal to or higher than the second predetermined value, the oil fluid L can be allowed to flow at a high flow rate through the first passage **92** when the main valve **91** opens. Thereby, a flow rate flowing through the passage between the sub-valve **110** and the valve seat part **139** is reduced. Therefore, for example, a rate of increase in damping force with respect to increase in the piston speed when the piston speed is in the normal speed region (equal to or higher than the second predetermined value) can be reduced. In other words, a slope of the rate of increase in the extension-side damping force with respect to the increase in the piston speed in the normal speed region (equal to or higher than the second predetermined value) can be laid down more than that in the very low speed region (lower than the second predetermined value). Thereby, the degree of freedom in design can be increased.
(150) In the extension stroke at the time of a high-frequency input (at the time of a small amplitude of vibration) in which a frequency higher than that at the time of the low-frequency input described above is input to the shock absorber **1**, an amount of the oil fluid L flowing from the upper chamber **22** to the upper chamber communication chamber **147** via the intermediate chamber **150** is small. Therefore, deformation of the flexible disc **100** is small, the upper chamber volume variable mechanism **186** can absorb a volume of the oil fluid L flowing into the upper chamber communication chamber **147** by an amount of bending of the flexible disc **100**, and a pressure rise in the upper chamber communication chamber **147** is reduced. Therefore, when the very low speed damping force rises, it is as if there is no flexible disc **100** and can be made in a state in which the upper chamber communication chamber **147** is in constant communication with the lower chamber **23** through the communication passage **148** of the cap member main body **95**, that is, the same state as a structure without the second damping force generation mechanism **183**.
(151) Therefore, in the extension stroke at the time of the high-frequency input, the rise of the very low speed damping force becomes gentler relative to damping force characteristics at the time of the low-frequency input. Also, in the extension stroke, in the very low speed region (lower than the second predetermined value), since the flexible disc **100** bends to open the second damping force generation mechanism **183** while increasing a volume of the oil fluid L flowing into the upper chamber communication chamber **147**, the very low speed damping force for the same piston speed has a characteristic in which it is lowered than that at the time of the low-frequency input in which the flexible disc **100** is fully bent and the volume of the oil fluid L flowing into the upper chamber communication chamber **147** remains unchanged. In other words, when a frequency of the piston **21** exceeds a predetermined frequency in the extension stroke, a flow rate of the oil fluid L to the sub-valve **110** of the second damping force generation mechanism **183** is restricted by the upper chamber volume variable mechanism **186** including the flexible disc **100**. Further, a change in damping force (inclination of a damping force with respect to a piston speed) until the second damping force generation mechanism **183** opens can be adjusted by a difference in rigidity (plate thickness or the like) of the flexible disc **100**.
(152) The main valve **71** of the first damping force generation mechanism **42** has higher rigidity and a higher valve opening pressure than the sub-valve **107** of the second damping force generation mechanism **173**. Therefore, in the compression stroke, in the very low speed region in which the piston speed is lower than the predetermined value, the second damping force generation

mechanism **173** opens while the first damping force generation mechanism **42** is in a closed state. In the normal speed region in which the piston speed is equal to or higher than the predetermined value, both the first damping force generation mechanism **42** and the second damping force generation mechanism **173** open. The sub-valve **107** is a very low speed valve that opens in a region in which the piston speed is very low to generate a damping force.

(153) That is, in the compression stroke, a pressure in the lower chamber **23** increases and a pressure in the upper chamber **22** decreases when the piston **21** moves to the lower chamber **23** side. Then, although none of the first damping force generation mechanisms **41** and **42** and the second damping force generation mechanisms **173** and **183** has a fixed orifice that allows constant communication between the lower chamber **23** and the upper chamber **22**, the oil fluid L in the lower chamber **23** flows into the lower chamber communication chamber **149** through the communication passages **148** of the cap member main body **95**. Thereby, the lower chamber communication chamber **149** rises in pressure. Therefore, in the lower chamber volume variable mechanism **185**, a portion of the flexible disc **100** on a radially inner side of a position in contact with the outer curved plate part **314** of the disc spring **117** bends to the spring contact disc **104** side to increase a volume of the lower chamber communication chamber **149**. Thereby, the lower chamber volume variable mechanism **185** suppresses a pressure rise in the lower chamber communication chamber **149**. At this time, since the flexible disc **100** bends and moves to the spring contact disc **104** side, the upper chamber volume variable mechanism **186** reduces a volume of the upper chamber communication chamber **147**.

(154) Here, in the compression stroke of the shock absorber **1** at the time of the low-frequency input (at the time of a large amplitude of vibration), since an amount of the oil fluid L flowing from the lower chamber **23** to the lower chamber communication chamber **149** as described above becomes large, the flexible disc **100** is greatly deformed. If an amount of deformation of the flexible disc **100** becomes large, a reaction force due to support stiffness of the clamped inner circumferential side increases to restrict the amount of deformation, and the lower chamber communication chamber **149** rises in pressure. As a result, the second passage **172** rises in pressure until reaching a state in which the second damping force generation mechanism **173** opens.

(155) At this time, since none of the first damping force generation mechanisms **41** and **42** and the second damping force generation mechanisms **173** and **183** has a fixed orifice that allows constant communication between the lower chamber **23** and the upper chamber **22**, a damping force rises sharply in the compression stroke in which the piston speed is lower than a third predetermined value at which the second damping force generation mechanism **173** opens. Also, in a very low speed region in which the piston speed is in a region higher than the third predetermined value and is lower than a fourth predetermined value that is a higher speed than the third predetermined value, the second damping force generation mechanism **173** opens while the first damping force generation mechanism **42** is in a closed state.

(156) That is, the sub-valve **107** is separated from the valve seat part **135** to allow communication between the lower chamber **23** and the upper chamber **22** through the compression-side second passage **172**. Therefore, the oil fluid L in the lower chamber **23** flows to the upper chamber **22** through the second passage part **162** in the valve seat member **109**, the passage between the sub-valve **107** and the valve seat part **135**, the intermediate chamber **150** and the upper chamber communication chamber **147**, the radial passage **222** in the passage groove **221** of the valve seat member **109**, the passage in the large diameter hole portion **133** of the valve seat member **109**, the piston rod passage part **51** in the passage notch part **30** of the piston rod **25**, the passage in the large diameter hole portion **46** of the piston **21**, the orifice **175**, and the passages in the plurality of passage holes **38** and the annular groove **55** of the piston **21**. Thereby, even in the very low speed region in which the piston speed is lower than the fourth predetermined value, a damping force due to valve characteristics (characteristics in which a damping force is substantially proportional to a piston speed) can be obtained.

(157) Also, in the compression stroke, in a normal speed region in which the piston speed is equal to or higher than the fourth predetermined value, the first damping force generation mechanism **42** opens while the second damping force generation mechanism **173** remains in an open state. That is, the sub-valve **107** is separated from the valve seat part **135** to allow the oil fluid L to flow from the lower chamber **23** to the upper chamber **22** through the compression-side second passage **172**. At this time, when the flow of the oil fluid L is reduced by the orifice **175** provided downstream of the sub-valve **107** in the second passage **172**, a pressure applied to the main valve **71** becomes high to increase a differential pressure, and the main valve **71** is separated from the valve seat part **50** to allow the oil fluid L to flow from the lower chamber **23** to the upper chamber **22** through the compression-side first passage **72**. Therefore, the oil fluid L in the lower chamber **23** flows to the upper chamber **22** through the passages in the plurality of passage holes **39** and the annular groove **56** and the passage between the main valve **71** and the valve seat part **50**.
(158) Thereby, even in the normal speed region in which the piston speed is equal to or higher than the fourth predetermined value, a damping force due to valve characteristics (a damping force is substantially proportional to a piston speed) can be obtained. A rate of increase in the compression-side damping force with respect to an increase in the piston speed in the normal speed region is lower than a rate of increase in the compression-side damping force with respect to an increase in the piston speed in the very low speed region. In other words, a slope of the rate of increase in the extension-side damping force with respect to the increase in the piston speed in the normal speed region can be laid down more than that in the very low speed region.
(159) Here, in the compression stroke, in the normal speed region in which the piston speed is equal to or higher than the fourth predetermined value, a differential pressure between the lower chamber **23** and the upper chamber **22** is larger than that in the low speed region in which the piston speed is equal to or higher than the third predetermined value and lower than the fourth predetermined value, but since the first passage **72** is not narrowed by an orifice, the oil fluid L can be allowed to flow at a high flow rate through the first passage **72** when the main valve **71** opens. In addition to this, when the second passage **172** is narrowed by the orifice **175**, deformation of the sub-valve **107** can be suppressed.
(160) At this time, the sub-valve **110** in a closed state receives a pressure in an opposite direction from the lower chamber **23** and the intermediate chamber **150**. Even if the differential pressure between the lower chamber **23** and the upper chamber **22** increases, since the orifice **175** is formed downstream of the sub-valve **110** in the second passage **172**, a pressure rise in the intermediate chamber **150** becomes gentle with respect to a pressure rise in the lower chamber **23**, and the pressure difference between the lower chamber **23** and the intermediate chamber **150** is inhibited from increasing. Therefore, the pressure difference between the lower chamber **23** and the intermediate chamber **150** received by the sub-valve **110** in a closed state is inhibited from increasing, and a large back pressure from the lower chamber **23** side toward the intermediate chamber **150** side can be inhibited from being applied to the sub-valve **110**.
(161) In the shock absorber **1**, the flow path for allowing the oil fluid L to flow from the lower chamber **23** to the upper chamber **22** in the compression stroke is provided in parallel by the first passage **72** and the second passage **172**, and the main valve **71** and the sub-valve **107** are provided in parallel. Also, the orifice **175** is connected in series to the sub-valve **107**.
(162) As described above, in the compression stroke, in the normal speed region in which the piston speed is equal to or higher than the fourth predetermined value, the oil fluid L can be allowed to flow at a high flow rate through the first passage **72** when the main valve **71** opens. Thereby, a flow rate flowing through the passage between the sub-valve **107** and the valve seat part **135** is reduced. Therefore, for example, reducing a rate of increase in damping force with respect to increase in the piston speed when the piston speed is in the normal speed region (equal to or higher than the fourth predetermined value) or the like is possible. In other words, a slope of the rate of increase in the compression-side damping force with respect to the increase in piston speed in the

normal speed region (equal to or higher than the fourth predetermined value) can be laid down more than that in the very low speed region (lower than the fourth predetermined value). Thereby, the degree of freedom in design can be increased.
(163) In the compression stroke at the time of the high-frequency input (at the time of a small amplitude of vibration) in which a frequency higher than that at the time of the low-frequency input described above is input to the shock absorber **1**, an amount of the oil fluid L flowing from the lower chamber **23** to the lower chamber communication chamber **149** is small.
(164) Therefore, deformation of the flexible disc **100** is small. Therefore, deformation of the flexible disc **100** is small, the lower chamber volume variable mechanism **185** can absorb a volume of the oil fluid L flowing into the lower chamber communication chamber **149** by an amount of bending of the flexible disc **100**, and a pressure rise in the lower chamber communication chamber **149** is reduced. Therefore, when the very low speed damping force rises, it is as if there is no flexible disc **100** and can be made in a state in which the lower chamber communication chamber **149** is in constant communication with the upper chamber **22** through the communication passage **302** of the spring contact disc **104**, that is, the same state as a structure without the second damping force generation mechanism **173**.
(165) Therefore, in the compression stroke at the time of the high-frequency input, the rise of the very low speed damping force becomes gentler relative to the damping force characteristics at the time of the low-frequency input. Also, in the compression stroke, in the very low speed region (lower than the fourth predetermined value), since the flexible disc **100** bends to open the second damping force generation mechanism **173** while increasing a volume of the oil fluid L flowing into the lower chamber communication chamber **149**, the very low speed damping force for the same piston speed has a characteristic in which it is lowered than that at the time of the low-frequency input in which the flexible disc **100** is fully bent and the volume of the oil fluid L flowing into the lower chamber communication chamber **149** remains unchanged. In other words, when the frequency of the piston **21** exceeds a predetermined frequency, a flow rate of the oil fluid L to the sub-valve **107** of the second damping force generation mechanism **173** is restricted by the lower chamber volume variable mechanism **185** including the flexible disc **100**. Further, a change in damping force (inclination of a damping force with respect to a piston speed) until the second damping force generation mechanism **173** opens can be adjusted by a difference in rigidity (plate thickness or the like) of the flexible disc **100**.
(166) Here, the compression stroke has characteristics in which the damping force characteristics due to the valve mechanism **255** are also combined.
(167) Patent Document 1 described above describes a shock absorber having two valves that open in the same stroke. When a structure having two valves that open in the same stroke is employed in this way, one valve can be opened in a region of a lower piston speed than the other valve, and both valves can be opened in a region of a higher speed than that. In a shock absorber having such a structure, if a damping force is set to be generated at the time of the low-frequency input in the very low speed region particularly to improve responsiveness at the time of a slight steering input or to improve ride comfort with a feeling of flatness on a smooth road, abnormal noise may occur at the time of the high-frequency input.
(168) In the shock absorber **1** of the first embodiment, the communication passages **148** that allow the inside and outside of the cap member **305** to communicate are formed in the bottom part **122** on one end side of the cap member **305** in the axial direction. The disc spring **116** is provided so that one end surface side in the axial direction is in contact with an outer circumferential side of the cap member **305** with respect to the communication passages **148**. The bendable flexible disc **100** is provided to be in contact with the other end surface side of the disc spring **116** in the axial direction. Therefore, the lower chamber communication chamber **149** communicating with the lower chamber **23** via the communication passages **148** can be formed between the cap member **305**, the flexible disc **100**, and the disc spring **116**. Then, the flexible disc **100** constitutes the lower

chamber volume variable mechanism **185** that changes a volume of the lower chamber communication chamber **149** by deforming.

(169) Thereby, in the compression stroke as described above, even if a damping force is set to be generated at the time of the low-frequency input in the very low speed region, a volume of the lower chamber communication chamber **149** provided in parallel with the second passage **172** can be changed by the lower chamber volume variable mechanism **185** at the time of the high-frequency input. Therefore, a flow rate of the oil fluid L flowing through the second passage **172** can be changed at the time of the high-frequency input. Therefore, in the compression stroke, even if a damping force is set to be generated at the time of the low-frequency input in the very low speed region, generation of abnormal noise during the compression stroke at the time of the high-frequency input can be suppressed.

(170) Also, in the shock absorber **1** of the first embodiment, the communication passages **302** that allow the inside and outside of the cap member **305** to communicate are formed in the spring contact disc **104** that is a bottom part of the cap member **305** on the other end side in the axial direction. The disc spring **117** is provided so that one end surface side in the axial direction is in contact with an outer circumferential side of the cap member **305** with respect to the communication passages **302**. The bendable flexible disc **100** is provided to be in contact with the other end surface side of the disc spring **117** in the axial direction. Therefore, the upper chamber communication chamber **147** communicating with the upper chamber **22** via the communication passages **302** can be formed between the cap member **305**, the flexible disc **100**, and the disc spring **117**. Then, the flexible disc **100** constitutes the upper chamber volume variable mechanism **186** that changes a volume of the upper chamber communication chamber **147** by deforming.

(171) Thereby, in the extension stroke as described above, even if a damping force is set to be generated at the time of the low-frequency input in the very low speed region, a volume of the upper chamber communication chamber **147** provided in the second passage **182** can be changed at the time of the high-frequency input by the upper chamber volume variable mechanism **186**. Therefore, a flow rate of the oil fluid L flowing through the second passage **182** can be changed at the time of the high-frequency input. Therefore, in the extension stroke, even if a damping force is set to be generated at the time of the low-frequency input in the very low speed region, generation of abnormal noise at the time of the high-frequency input can be suppressed.

(172) Also, the disc spring **116** and the disc spring **117** are provided on both one end surface side and the other end surface side of the flexible disc **100** in the axial direction. Therefore, the lower chamber volume variable mechanism **185** including the lower chamber communication chamber **149** and the upper chamber volume variable mechanism **186** including the upper chamber communication chamber **147** can be provided. Therefore, in both the extension stroke and the compression stroke, even if a damping force is set to be generated at the time of the low-frequency input in the very low speed region, generation of abnormal noise at the time of the high-frequency input can be suppressed as described above.

(173) Also, the disc spring **116** has a plate shape, and is formed so that the outer curved plate part **314** on the outer circumferential surface side is bent to be in contact with the flexible disc **100** and the inner curved plate part **312** on the inner circumferential surface side is bent to be in contact with the bottom part **122** of the cap member **305**. Therefore, compared to a disc spring that is in contact with the flexible disc **100** and the bottom part **122** at end portions of inner and outer circumferences thereof, leakage of the oil fluid L from the lower chamber communication chamber **149** can be suppressed. Therefore, damping force characteristics can be stabilized. That is, since the end portions of the inner and outer circumferences of the disc spring are formed by press forming, burrs, scratches, or the like are likely to occur. Therefore, the disc spring that is in contact with the flexible disc **100** and the bottom part **122** at the end portions of the inner and outer circumferences is likely to leak the oil fluid L, but since the disc spring **116** is not in contact with the flexible disc **100** and the bottom part **122** at its end portions of inner and outer circumferences, such leakage can

be suppressed.

(174) The disc spring **117** has a plate shape, and is formed so that the outer curved plate part **314** on the outer circumferential surface side is bent to be in contact with the flexible disc **100** and the inner curved plate part **312** on the inner circumferential surface side is bent to be in contact with the spring contact disc **104** of the cap member **305**. Therefore, compared to a disc spring that is in contact with the flexible disc **100** and the spring contact disc **104** at end portions of inner and outer circumferences thereof, leakage of the oil fluid L from the upper chamber communication chamber **147** can be suppressed. Therefore, damping force characteristics can be stabilized.

(175) The disc springs **116** and **117** come into contact with the cylindrical part **124** of the cap member **305** that covers the valve seat member **109** at their outer circumferential end portions. Therefore, radial displacement of the disc springs **116** and **117** with respect to the flexible disc **100** can be suppressed by the cylindrical part **124**. Therefore, leakage of the oil fluid from the lower chamber communication chamber **149** due to radial displacement of the disc spring **116** with respect to the flexible disc **100**, and leakage of the oil fluid from the upper chamber communication chamber **147** due to radial displacement of the disc spring **117** with respect to the flexible disc **100** can be suppressed. Therefore, damping force characteristics can be further stabilized. Also, since there is no need to provide a dedicated part for suppressing radial displacement of the disc springs **116** and **117** with respect to the flexible disc **100**, the number of parts can be reduced and costs can be reduced.

Second Embodiment

(176) Next, a second embodiment will be described mainly on the basis of FIG. **4**, focusing on differences from the first embodiment. Further, parts common to those in the first embodiment are denoted by the same terms and the same reference signs.

(177) In a shock absorber **1**A of the second embodiment, an accumulator **190**A is provided as illustrated in FIG. **4** in place of the accumulator **190** of the first embodiment. The accumulator **190**A includes a disc **97**, a disc **98**, a disc **99**, and a flexible disc **100** stacked on a cylindrical part **124** side in an axial direction of the bottom part **122** of the cap member main body **95** in order from a bottom part **122** side. Only one disc **98** is provided in the accumulator **190**A. The accumulator **190**A includes a disc spring **116** between the bottom part **122** and the flexible disc **100** of the cap member main body **95**.

(178) On the other hand, the accumulator **190**A does not include the disc spring **117** of the first embodiment. Also, the accumulator **190**A does not include the discs **101** to **103** of the first embodiment, but includes one disc **401**A instead. Further, the shock absorber **1**A includes a disc **402**A in place of the spring contact disc **104**. In the shock absorber **1**A, the cap member main body **95** serves as a cap member.

(179) The disc **401**A is in contact with the flexible disc **100** on a side of the flexible disc **100** opposite to the disc **99**. The disc **401**A has a bored disc shape made of a metal. The disc **401**A is formed to have the same inner diameter and the same outer diameter as the disc **99**, and has a thickness larger than that of the disc **99**. A mounting shaft part **28** of a piston rod **25** is fitted on an inner circumferential side of the disc **401**A. Thereby, the disc **401**A is radially positioned with respect to the piston rod **25**.

(180) The disc **402**A is in contact with the disc **401**A on a side of the disc **401**A opposite to the flexible disc **100**. The disc **402**A has a bored disc shape made of a metal. The disc **402**A is formed to have the same inner diameter and the same outer diameter as a sub-valve **107**. The disc **402**A is formed to have a larger thickness and higher rigidity than the flexible disc **100**. A spring member **105** is in contact with a base plate part **331** on a side of the disc **402**A opposite to the disc **401**A. The mounting shaft part **28** of the piston rod **25** is fitted on an inner circumferential side of the disc **402**A. Thereby, the disc **402**A is radially positioned with respect to the piston rod **25**.

(181) Also in the accumulator **190**A, a lower chamber communication chamber **149** is formed by being surrounded by the flexible disc **100**, the disc spring **116**, the discs **97** to **99**, and the bottom

part **122** of the cap member main body **95**. In the accumulator **190**A, the flexible disc **100**, the disc spring **116**, the discs **97** to **99**, and the lower chamber communication chamber **149** constitute a lower chamber volume variable mechanism **185**A capable of changing a volume of the lower chamber communication chamber **149**.

(182) The shock absorber **1**A includes an intermediate chamber **150**A between the flexible disc **100** and a valve seat member **109**, and the upper chamber communication chamber **147** of the first embodiment is not provided. Similarly to the intermediate chamber **150** of the first embodiment, the intermediate chamber **150**A is in constant communication with a radial passage **222** and thus is in constant communication with an upper chamber **22** (see FIG. **2**). In the accumulator **190**A, the flexible disc **100** and the disc spring **116** constitute an upper chamber volume variable mechanism **186**A that changes a volume of the upper chamber **22** (see FIG. **2**) by changing a volume of the intermediate chamber **150**A.

(183) In the shock absorber **1**A of the second embodiment, in an extension stroke, the flexible disc **100** constituting the upper chamber volume variable mechanism **186**A deforms a portion of the disc spring **116** on a radially inner side of a portion in contact with an outer curved plate part **314** toward the bottom part **122** side, and thereby a volume of the intermediate chamber **150**A is increased. Thereby, in the extension stroke, the accumulator **190**A operates in substantially the same manner as in the first embodiment.

(184) On the other hand, in the accumulator **190**A, an oil fluid L enters the lower chamber communication chamber **149** of the lower chamber volume variable mechanism **185**A from a lower chamber **23** through a communication passage **148** in a compression stroke.

(185) As a pressure in the lower chamber communication chamber **149** increases, the flexible disc **100** deforms to the disc **402**A side, the disc spring **116** also deforms following the deformation to keep the outer curved plate part **314** in a state of being in contact with the flexible disc **100**, and thereby a volume of the lower chamber communication chamber **149** is increased. Thereby, the accumulator **190**A operates in substantially the same manner as in the first embodiment also in the compression stroke.

(186) According to the shock absorber **1**A of the second embodiment, in a structure in which the accumulator **190**A supports the flexible disc **100** with one disc spring **116** on one side, a damping force can be made variable according to a frequency in both an extension process and a compression process, and thereby the structure can be simplified.

Third Embodiment

(187) Next, a third embodiment will be described mainly on the basis of FIGS. **5** and **6**, focusing on differences from the first embodiment. Further, parts common to those in the first embodiment are denoted by the same terms and the same reference signs.

(188) In the shock absorber **1**B of the third embodiment, a cap member **305**B is provided as illustrated in FIG. **5** in place of the cap member **305** of the first embodiment. The cap member **305**B includes a cap member main body **95**B partially different from the cap member main body **95** of the first embodiment, a spring contact disc **104** similar to that of the first embodiment, and a planar disc **411**B that is not provided in the first embodiment.

(189) The cap member main body **95**B includes an intermediate tapered part **123**B formed between a bottom part **122** similar to that of the first embodiment and a cylindrical part **124** similar to that of the first embodiment in place of the intermediate curved part **123** of the first embodiment. The intermediate tapered part **123**B has a tapered shape coaxial with the bottom part **122**. The intermediate tapered part **123**B extends from an outer circumferential edge portion of the bottom part **122** while a diameter thereof increases to one side of the bottom part **122** in the axial direction to be connected to the cylindrical part **124**.

(190) The planar disc **411**B has a bored disc shape made of a metal. As illustrated in FIG. **6**, the planar disc **411**B includes an inner annular part **412**B, an outer annular part **413**B, and a plurality, specifically two, of connection parts **414**B connecting them. The inner annular part **412**B has a

bored disc shape. The outer annular part **413**B has a bored disc shape having an inner diameter larger than an outer diameter of the inner annular part **412**B.
(191) Both the inner annular part **412**B and the outer annular part **413**B have a constant width in a radial direction over the entire circumference. The outer annular part **413**B has a radial width larger than a radial width of the inner annular part **412**B. The inner annular part **412**B and the outer annular part **413**B are coaxially disposed. The two connection parts **414**B connect the inner annular part **412**B and the outer annular part **413**B in a state in which they are coaxial. The two connection parts **414**B connect an outer circumferential edge portion of the inner annular part **412**B and an inner circumferential edge portion of the outer annular part **413**B. The two connection parts **414**B are provided to be shifted in position by 180° in a circumferential direction of the inner annular part **412**B and the outer annular part **413**B.
(192) With a shape described above, the planar disc **411**B includes two arcuate passage holes **126**B formed to be surrounded by the inner annular part **412**B, the outer annular part **413**B, and the two connection parts **414**B. The two passage holes **126**B penetrate the planar disc **411**B in a thickness direction (axial direction).
(193) As illustrated in FIG. **5**, the outer annular part **413**B of the planar disc **411**B has an outer diameter slightly smaller than a minimum inner diameter of the intermediate tapered part **123**B of the cap member main body **95**B. The planar disc **411**B is disposed between the bottom part **122** of the cap member main body **95**B, and the disc **97** and the disc spring **116**. The planar disc **411**B is provided so that one end surface in the axial direction is in contact with the bottom part **122** and the other end surface in the axial direction is in contact with the disc **97** and the disc spring **116**. The planar disc **411**B is axially clamped by the bottom part **122** of the cap member main body **95**B and the disc **97** at the inner annular part **412**B and portions of two connection parts **414**B on the inner annular part **412**B side.
(194) The planar disc **411**B is provided so that one end surface in the axial direction is in contact with the bottom part **122** and the other end surface in the axial direction is in contact with an inner curved plate part **312** of the disc spring **116**. The planar disc **411**B, together with the bottom part **122**, constitutes a bottom part **122**B of the cap member **305**B by overlapping the bottom part **122**. Thus, the bottom part **122**B of the cap member **305**B includes the planar disc **411**B that is in contact with the disc spring **116**.
(195) The planar disc **411**B is radially positioned with respect to a piston rod **25** by fitting a mounting shaft part **28** of the piston rod **25** to an inner circumferential side of the inner annular part **412**B. In the planar disc **411**B, a communication passage **148**B in each of the passage holes **126**B is in constant communication with a communication passage **148** of the bottom part **122**. Therefore, the communication passage **148**B is in constant communication with a lower chamber **23**.
(196) Even if the disc spring **116** moves to the maximum in the radial direction, the disc spring **116** maintains a state in which the inner curved plate part **312** is in contact with a position of the bottom part **122**B on a radially outer side of all the passage holes **126**B of the bottom part **122**B over the entire circumference. The disc spring **116** is provided so that one end surface side in the axial direction is in contact with an outer circumferential side of the communication passage **148**B of the cap member **305**B.
(197) The shock absorber **1**B of the third embodiment operates in the same manner as the shock absorber **1** of the first embodiment except that the lower chamber communication chamber **149** allows an oil fluid L to flow to the lower chamber **23** via the communication passage **148**B and the communication passage **148**.
(198) According to the shock absorber **1**B of the third embodiment, the bottom part **122**B of the cap member **305**B includes the planar disc **411**B that is in contact with the inner curved plate part **312** of the disc spring **116**. Therefore, flatness of a portion of the bottom part **122**B in contact with the disc spring **116** can be improved. Therefore, leakage of the oil fluid L from the lower chamber communication chamber **149** via a space between the disc spring **116** and the bottom part **122**B can

be further suppressed. Therefore, damping force characteristics can be further stabilized.
(199) Further, it is also possible to have a structure without the disc spring **117** when the same modification as the modification of the shock absorber **1**A of the second embodiment with respect to the shock absorber **1** of the first embodiment is performed for the shock absorber **1**B of the third embodiment.
Fourth Embodiment
(200) Next, a fourth embodiment will be described mainly on the basis of FIGS. **7** and **8**, focusing on differences from the third embodiment. Further, parts common to those in the third embodiment are denoted by the same terms and the same reference signs.
(201) In a shock absorber **1**C of the fourth embodiment, a cap member **305**C is provided as illustrated in FIG. **7** in place of the cap member **305**B of the third embodiment. The cap member **305**C includes a cap member main body **95**B similar to that of the third embodiment and a spring contact disc **104** similar to that of the third embodiment. The cap member **305**C includes a planar disc **411**C in place of the planar disc **411**B of the third embodiment.
(202) The planar disc **411**C is constituted by two parts including an inner disc **412**C having a bored disc shape made of a metal, and an outer disc **413**C having a bored disc shape as illustrated in FIG. **8** and formed of a rubber material having a sealing property such as nitrile rubber or fluoro-rubber. As illustrated in FIG. **7**, the outer disc **413**C has an inner diameter larger than an outer diameter of the inner disc **412**C. Both the inner disc **412**C and the outer disc **413**C have a constant width in a radial direction over the entire circumference. The outer disc **413**C has a radial width larger than a radial width of the inner disc **412**C. The inner disc **412**C and the outer disc **413**C have the same thickness.
(203) The planar disc **411**C has a circular passage hole **126**C between the inner disc **412**C and the outer disc **413**C. The passage hole **126**C penetrates the planar disc **411**C in a thickness direction (axial direction).
(204) The outer diameter of the outer disc **413**C of the planar disc **411**C has an outer diameter slightly smaller than the minimum inner diameter of the intermediate tapered part **123**B of the cap member main body **95**B. The planar disc **411**C is disposed between a bottom part **122** of the cap member main body **95**B, and a disc **97** and a disc spring **116**. The inner disc **412**C is in contact with the bottom part **122** of the cap member main body **95**B and the disc **97**. The outer disc **413**C is in contact with the bottom part **122** of the cap member main body **95**B and the disc spring **116**. The inner disc **412**C is provided so that one end surface in the axial direction is in contact with the bottom part **122** and the other end surface in the axial direction is in contact with the disc **97**. The inner disc **412**C is axially clamped by the bottom part **122** of the cap member main body **95**B and the disc **97**. The outer disc **413**C is provided so that one end surface in the axial direction is in contact with the bottom part **122** and the other end surface in the axial direction is in contact with an inner curved plate part **312** of the disc spring **116**. A state of the outer disc **413**C being in contact with the bottom part **122** due to a biasing force of the disc spring **116** is maintained.
(205) The planar disc **411**C, together with the bottom part **122**, constitutes a bottom part **122**C of the cap member **305**C by the inner disc **412**C and the outer disc **413**C overlapping the bottom part **122**. Thus, the bottom part **122**C of the cap member **305**C has the planar disc **411**C including the outer disc **413**C that is in contact with the disc spring **116**.
(206) In the planar disc **411**C, the inner disc **412**C is radially positioned with respect to a piston rod **25** by fitting a mounting shaft part **28** of the piston rod **25** to an inner circumferential side of the inner disc **412**C. Also, in the planar disc **411**C, the outer disc **413**C is radially positioned with respect to the cap member main body **95**C, that is, the piston rod **25**, by the intermediate tapered part **123**B of the cap member main body **95**C. In the planar disc **411**C, a communication passage **148**C in the passage hole **126**C is in constant communication with a communication passage **148** of the bottom part **122**.
(207) Even if the disc spring **116** moves to the maximum in the radial direction, the disc spring **116**

maintains a state in which the inner curved plate part **312** is in contact with a position of the bottom part **122**C on a radially outer side of the passage hole **126**C of the bottom part **122**C over the entire circumference. The disc spring **116** is provided so that one end surface side in the axial direction is in contact with an outer circumferential side of the communication passage **148**C of the cap member **305**C.
(208) The shock absorber **1**C of the fourth embodiment operates in the same manner as the shock absorber **1** of the first embodiment except that a lower chamber communication chamber **149** allows an oil fluid L to flow to the lower chamber **23** via the communication passage **148**C and the communication passage **148**.
(209) According to the shock absorber **1**C of the fourth embodiment, the bottom part **122**C of the cap member **305**C includes the outer disc **413**C having a high sealing property that is in contact with the inner curved plate part **312** of the disc spring **116**. Therefore, leakage of the oil fluid L from the lower chamber communication chamber **149** through a space between the disc spring **116** and the bottom part **122**C can be further suppressed. Therefore, damping force characteristics can be further stabilized.
(210) Further, it is also possible to have a structure without the disc spring **117** when the same modification as the modification of the shock absorber **1**A of the second embodiment with respect to the shock absorber **1** of the first embodiment is performed for the shock absorber **1**C of the fourth embodiment.
Fifth Embodiment
(211) Next, a fifth embodiment will be described mainly on the basis of FIGS. **9** and **10**, focusing on differences from the first embodiment. Further, parts common to those in the first embodiment are denoted by the same terms and the same reference signs.
(212) In a shock absorber **1**D of the fifth embodiment, an accumulator **190**D is provided as illustrated in FIG. **9** in place of the accumulator **190** of the first embodiment. The accumulator **190**D is not provided inside a cap member main body **95**, but is provided outside the cap member main body **95**. Also, in the shock absorber **1**D, a spring contact disc **104** similar to that of the first embodiment also is not provided inside a cap member main body **95**, but is provided outside the cap member main body **95**. The accumulator **190**D and the spring contact disc **104** are provided between the bottom part **122** of the cap member main body **95** and a disc **87**. In the shock absorber **1**D, the cap member main body **95** serves as a cap member.
(213) In the shock absorber **1**D, the spring contact disc **104** is in contact with the disc **87** on a side opposite to a disc **86**. The accumulator **190**D is provided on a side of the spring contact disc **104** opposite to the disc **87**. The accumulator **190**D includes a disc **97**, a disc **98**, a guide member **431**D, a disc **99**, a flexible disc **100**, a disc **101**, a guide member **432**D, a disc **102**, and a disc **103** in order from the spring contact disc **104** side. Then, in the accumulator **190**D, a side of the disc **103** opposite to the disc **102** is in contact with the bottom part **122** of the cap member main body **95**.
(214) The accumulator **190**D includes a disc spring **116** disposed between the spring contact disc **104** and the flexible disc **100**, and a disc spring **117** disposed between the flexible disc **100** and the bottom part **122** of the cap member main body **95**.
(215) The guide member **431**D has a flat plate shape made of a metal. As illustrated in FIG. **10**, the guide member **431**D has a substantially equilateral triangular shape with a hole formed in a center thereof. The guide member **431**D has an arcuate guide part **435**D at a position of each vertex of the triangle at the outer circumferential portion. The guide member **431**D includes three guide parts **435**D at regular intervals in a circumferential direction.
(216) As illustrated in FIG. **9**, the guide member **431**D is radially positioned with respect to a piston rod **25** by fitting a mounting shaft part **28** of the piston rod **25** to an inner circumferential side of the guide member **431**D. In this state, the inner circumferential side of the guide member **431**D is axially clamped by the disc **98** and the disc **99**. Three guide parts **435**D of the guide member **431**D are in contact with an inner circumferential side of an intermediate tapered plate part

**313** of the disc spring **116** from a radially inner side at the same time. Thereby, the disc spring **116** is radially positioned with respect to the piston rod **25**.
(217) The guide member **432**D is a common part having the same shape as the guide member **431**D. The guide member **432**D is also radially positioned with respect to the piston rod **25** by fitting the mounting shaft part **28** of the piston rod **25** to an inner circumferential side of the guide member **432**D. In this state, the guide member **432**D is axially clamped by the disc **101** and the disc **102**. Three guide parts **435**D of the guide member **432**D are in contact with an inner circumferential side of the intermediate tapered plate part **313** of the disc spring **117** from a radially inner side at the same time. Thereby, the disc spring **117** is radially positioned with respect to the piston rod **25**.
(218) In the shock absorber **1**D, a lower chamber communication chamber **149** is formed by being surrounded by the flexible disc **100**, the disc spring **116**, the discs **97** to **99**, the guide member **431**D, and the spring contact disc **104**. The lower chamber communication chamber **149** communicates with a lower chamber **23** via a communication passage **302** of the spring contact disc **104**.
(219) The guide member **431**D is disposed between the disc spring **116** and the flexible disc **100**. The guide member **431**D is provided inside the lower chamber communication chamber **149**. The flexible disc **100**, the disc springs **116** and **117**, the discs **97** to **99**, the guide member **431**D, and the lower chamber communication chamber **149** constitute a lower chamber volume variable mechanism **185**D capable of changing a volume of the lower chamber communication chamber **149**.
(220) In the shock absorber **1**D, an upper chamber communication chamber **147** is formed by being surrounded by the flexible disc **100**, the disc spring **117**, the discs **101** to **103**, the guide member **432**D, and the bottom part **122** of the cap member main body **95**.
(221) In the shock absorber **1**D, the accumulator **190**D and the spring contact disc **104** are provided outside the cap member main body **95**. Therefore, a spring member **105** is in contact with a base plate part **331** on an end surface in the axial direction of the bottom part **122** of the cap member main body **95** on a cylindrical part **124** side.
(222) In the shock absorber **1**D, a space between the bottom part **122** and a valve seat member **109** of the cap member main body **95** is an intermediate chamber **150** that is in constant communication with a radial passage **222**. The upper chamber communication chamber **147** is in constant communication with the intermediate chamber **150** via a communication passage **148** of the bottom part **122** of the cap member main body **95**. Therefore, the upper chamber communication chamber **147** is in constant communication with an upper chamber **22** (see FIG. **2**).
(223) The guide member **432**D is disposed between the disc spring **117** and the flexible disc **100** and provided inside the upper chamber communication chamber **147**. The flexible disc **100**, the disc springs **116** and **117**, the discs **101** to **103**, the guide member **432**D, and the upper chamber communication chamber **147** constitute an upper chamber volume variable mechanism **186**D capable of changing a volume of the upper chamber communication chamber **147**.
(224) In the shock absorber **1**D of the fifth embodiment, in an extension stroke, the flexible disc **100** constituting the upper chamber volume variable mechanism **186**D deforms a portion of the disc spring **116** on a radially inner side of a portion in contact with an outer curved plate part **314** toward the spring contact disc **104** side to increase a volume of the upper chamber communication chamber **147**. Also, in a compression stroke, the flexible disc **100** constituting the lower chamber volume variable mechanism **185**D deforms a portion of the disc spring **117** on a radially inner side of a portion in contact with the outer curved plate part **314** toward the bottom part **122** side of the cap member main body **95** to increase a volume of the lower chamber communication chamber **149**. Thereby, the accumulator **190**D operates in the same manner as in the first embodiment.
(225) According to the shock absorber **1**D of the fifth embodiment, the guide member **431**D is provided between the disc spring **116** and the flexible disc **100** to be in contact with an inner

circumferential side of the disc spring **116**, and the guide member **432**D is provided between the disc spring **117** and the flexible disc **100** to be in contact with an inner circumferential side of the disc spring **117**. Therefore, radial displacement of the disc springs **116** and **117** with respect to the flexible disc **100** can be suppressed by the guide members **431**D and **432**D. Therefore, leakage of the oil fluid from the lower chamber communication chamber **149** due to radial displacement of the disc spring **116** with respect to the flexible disc **100**, and leakage of the oil fluid from the upper chamber communication chamber **147** due to radial displacement of the disc spring **117** with respect to the flexible disc **100** can be suppressed. Therefore, damping force characteristics can be further stabilized.

(226) Further, it is also possible to have a structure without the disc spring **117** when the same modification as the modification of the shock absorber **1**A of the second embodiment with respect to the shock absorber **1** of the first embodiment is performed for the shock absorber **1**D of the fifth embodiment.

Sixth Embodiment

(227) Next, a sixth embodiment will be described mainly on the basis of FIG. **11**, focusing on differences from the first embodiment. Further, parts common to those in the first embodiment are denoted by the same terms and the same reference signs.

(228) In a shock absorber **1**E of the sixth embodiment, as illustrated in FIG. **11**, an accumulator **190**E is provided in place of the accumulator **190** of the first embodiment.

(229) The accumulator **190**E includes a seal disc **116**E (biasing member) having a bored disc shape provided on a radially outer side of discs **97** to **99** instead of the disc spring **116**. The accumulator **190**E includes a seal disc **117**E (biasing member) having a bored disc shape provided on a radially outer side of discs **101** to **103** instead of the disc spring **117**.

(230) The seal disc **116**E is provided between a bottom part **122** of a cap member main body **95** and a flexible disc **100**. The seal disc **116**E includes a base disc **441**E having a bored disc shape made of a metal, an annular inner circumferential seal part **312**E formed of a rubber material having a sealing property such as nitrile rubber or fluoro-rubber, and an annular outer circumferential seal part **314**E formed of a rubber material having a sealing property such as nitrile rubber or fluoro-rubber.

(231) The base disc **441**E has a constant width in a radial direction over the entire circumference. The inner circumferential seal part **312**E is fixed to an inner circumferential edge portion of the base disc **441**E over the entire circumference and has an annular shape. A cross section of the inner circumferential seal part **312**E in a plane including a central axis of the base disc **441**E has a triangular shape that protrudes to one side in a thickness direction (axial direction) of the base disc **441**E. An inner circumferential edge portion of the inner circumferential seal part **312**E is an inner circumferential edge portion of the seal disc **116**E, and a radially inner side of the inner circumferential seal part **312**E is a hole **310**E that penetrates the seal disc **116**E in the axial direction. An inner diameter of the inner circumferential seal part **312**E, that is, an inner diameter of the hole **310**E, is larger than an outer diameter of the disc **98**.

(232) The outer circumferential seal part **314**E is fixed to an outer circumferential edge portion of the base disc **441**E over the entire circumference and has an annular shape. A cross section of the outer circumferential seal part **314**E in a plane including the central axis of the base disc **441**E has a triangular shape that protrudes to the other side in the thickness direction (axial direction) of the base disc **441**E. That is, the inner circumferential seal part **312**E and the outer circumferential seal part **314**E have shapes that protrude in directions opposite to each other in the axial direction from the base disc **441**E. An outer circumferential edge portion of the outer circumferential seal part **314**E is an outer circumferential edge portion of the seal disc **116**E. The seal disc **116**E has an outer diameter slightly smaller than an inner diameter of a cylindrical part **124** of the cap member main body **95**.

(233) The seal disc **116**E is directed such that the inner circumferential seal part **312**E protrudes to

the bottom part **122** side of the cap member main body **95** and the outer circumferential seal part **314**E protrudes to the flexible disc **100** side from the base disc **441**E. The seal disc **116**E is in contact with the bottom part **122** at the inner circumferential seal part **312**E. The inner circumferential seal part **312**E is in contact with a position of the bottom part **122** on a radially outer side of all passage holes **126** of the bottom part **122** over the entire circumference. The seal disc **116**E is in contact with the flexible disc **100** over the entire circumference at the outer circumferential seal part **314**E.
(234) A range of movement of the seal disc **116**E in a radial direction of a piston rod is determined when the seal disc **116**E comes into contact with the cylindrical part **124** of the cap member main body **95**. In other words, in the radial direction of the piston rod **25**, the seal disc **116**E is slightly movable within an inner range of the cylindrical part **124** of the cap member main body **95** with respect to the cap member main body **95**, the discs **97** to **99**, and the flexible disc **100** which do not move. However, even if the seal disc **116**E moves to the maximum in the radial direction, the seal disc **116**E maintains a state in which the inner circumferential seal part **312**E is in contact with a position of the bottom part **122** on a radially outer side of all the passage holes **126** of the bottom part **122** over the entire circumference and maintains a state in which the outer circumferential seal part **314**E is in contact with the flexible disc **100** over the entire circumference. An outer circumferential end portion of the seal disc **116**E comes into contact with the cylindrical part **124** of the cap member main body **95** that covers a valve seat member **109**.
(235) The seal disc **117**E is provided between the flexible disc **100** and a spring contact disc **104**. The seal disc **117**E is a common part having the same shape as the seal disc **116**E. The seal disc **117**E is directed in a direction opposite to the seal disc **116**E in the axial direction. The discs **101** to **103** are disposed on an inner side of the seal disc **117**E in the radial direction with a gap with respect to an inner circumferential edge portion of disc spring **117**E in the radial direction.
(236) The seal disc **117**E is directed such that the inner circumferential seal part **312**E protrudes to the spring contact disc **104** side and the outer circumferential seal part **314**E protrudes to the flexible disc **100** side from the base disc **441**E. The seal disc **117**E is in contact with the spring contact disc **104** at the inner circumferential seal part **312**E. The inner circumferential seal part **312**E of the seal disc **117**E is in contact with a position of the bottom part **122** on a radially outer side of all passage holes **301** of the spring contact disc **104** over the entire circumference. The seal disc **117**E is in contact with the flexible disc **100** over the entire circumference at the outer circumferential seal part **314**E.
(237) A range of movement of the seal disc **117**E in the radial direction of the piston rod **25** is also determined when the seal disc **117**E comes into contact with the cylindrical part **124** of the cap member main body **95**. Even if the seal disc **117**E moves to the maximum in the radial direction, the seal disc **117**E maintains a state in which the inner circumferential seal part **312**E is in contact with a position of the spring contact disc **104** on a radially outer side of all the passage holes **301** of the spring contact disc **104** over the entire circumference and maintains a state in which the outer circumferential seal part **314**E is in contact with the flexible disc **100** over the entire circumference. An outer circumferential end portion of the seal disc **117**E comes into contact with the cylindrical part **124** of the cap member main body **95** that covers the valve seat member **109**.
(238) A lower chamber communication chamber **149** is formed by being surrounded by the discs **97** to **99**, the flexible disc **100**, the seal disc **116**E, and the bottom part **122** of the cap member main body **95**. The flexible disc **100**, the seal discs **116**E and **117**E, the discs **97** to **99**, and the lower chamber communication chamber **149** constitute a lower chamber volume variable mechanism **185**E capable of changing a volume of the lower chamber communication chamber **149**.
(239) An upper chamber communication chamber **147** is formed by being surrounded by the discs **101** to **103**, the flexible disc **100**, the seal disc **117**E, and the spring contact disc **104**. The flexible disc **100**, the seal discs **116**E and **117**E, the discs **101** to **103**, and the upper chamber communication chamber **147** constitute an upper chamber volume variable mechanism **186**E

capable of changing a volume of the upper chamber communication chamber **147**.
(240) In the shock absorber **1**E of the sixth embodiment, in an extension stroke, the flexible disc **100** constituting the upper chamber volume variable mechanism **186**E deforms a portion of the seal disc **116**E on a radially inner side of a portion in contact with the outer circumferential seal part **314**E toward the bottom part **122** side of the cap member main body **95** to increase a volume of the upper chamber communication chamber **147**. Also, in a compression stroke, the flexible disc **100** constituting the lower chamber volume variable mechanism **185**E deforms a portion of the seal disc **117**E on a radially inner side of a portion in contact with the outer circumferential seal part **314**E toward the spring contact disc **104** side to increase a volume of the lower chamber communication chamber **149**. Thereby, the accumulator **190**E operates in the same manner as the accumulator **190** in the first embodiment.
(241) According to the shock absorber **1**E of the sixth embodiment, the seal disc **116**E is in contact with the flexible disc **100** at the outer circumferential seal part **314**E made of a material with a high sealing property, and is in contact with the bottom part **122** of the cap member main body **95** at the inner circumferential seal part **312**E made of a material with a high sealing property. Therefore, leakage of an oil fluid L from the lower chamber communication chamber **149** can be suppressed. Further, the seal disc **117**E is in contact with the flexible disc **100** at the outer circumferential seal part **314**E made of a material with a high sealing property, and is in contact with the spring contact disc **104** at the inner circumferential seal part **312**E made of a material with a high sealing property. Therefore, leakage of the oil fluid L from the upper chamber communication chamber **147** can be suppressed. Therefore, damping force characteristics can be further stabilized.
(242) Further, in the shock absorber **1** of the first embodiment illustrated in FIGS. **1** to **3**, portions of the disc springs **116** and **117** in contact with the flexible disc **100** or portions of the flexible disc **100** in contact with the disc springs **116** and **117** may be coated with a sealing material. Also, in the first embodiment, a portion of the disc spring **116** in contact with the bottom part **122** of the cap member main body **95** or a portion of the bottom part **122** of the cap member main body **95** in contact with the disc spring **116** may be coated with a sealing material. Also, in the first embodiment, a portion of the disc spring **117** in contact with the spring contact disc **104** or a portion of the spring contact disc **104** in contact with the disc spring **117** may be coated with a sealing material.
(243) In the shock absorber **1**A of the second embodiment illustrated in FIG. **4**, a portion of the disc spring **116** in contact with the flexible disc **100** or a portion of the flexible disc **100** in contact with the disc spring **116** may be coated with a sealing material. Also, in the second embodiment, a portion of the disc spring **116** in contact with the bottom part **122** of the cap member main body **95** or a portion of the bottom part **122** of the cap member main body **95** in contact with the disc spring **116** may be coated with a sealing material.
(244) In the shock absorber **1**B of the third embodiment illustrated in FIG. **5**, portions of the disc springs **116** and **117** in contact with the flexible disc **100** or portions of the flexible disc **100** in contact with the disc springs **116** and **117** may be coated with a sealing material. Also, in the third embodiment, a portion of the disc spring **116** in contact with the planar disc **411**B or a portion of the planar disc **411**B in contact with the disc spring **116** may be coated with a sealing material. Also, in the third embodiment, a portion of the disc spring **117** in contact with the spring contact disc **104** or a portion of the spring contact disc **104** in contact with the disc spring **117** may be coated with a sealing material.
(245) In the shock absorber **1**C of the fourth embodiment illustrated in FIG. **7**, portions of the disc springs **116** and **117** in contact with the flexible disc **100** or portions of the flexible disc **100** in contact with the disc springs **116** and **117** may be coated with a sealing material. Also, in the fourth embodiment, a portion of the disc spring **117** in contact with the spring contact disc **104** or a portion of the spring contact disc **104** in contact with the disc spring **117** may be coated with a sealing material.

(246) In the shock absorber **1**D of the fifth embodiment illustrated in FIG. **9**, portions of the disc springs **116** and **117** in contact with the flexible disc **100** or portions of the flexible disc **100** in contact with the disc springs **116** and **117** may be coated with a sealing material. Also, in the fifth embodiment, a portion of the disc spring **116** in contact with the spring contact disc **104** or a portion of the spring contact disc **104** in contact with the disc spring **116** may be coated with a sealing material. Also, in the fifth embodiment, a portion of the disc spring **117** in contact with the bottom part **122** of the cap member main body **95** or a portion of the bottom part **122** of the cap member main body **95** in contact with the disc spring **117** may be coated with a sealing material.
(247) Further, it is also possible to have a structure without the disc spring **117**E when the same modification as the modification of the shock absorber **1**A of the second embodiment with respect to the shock absorber **1** of the first embodiment can be performed for the shock absorber **1**E of the sixth embodiment.
(248) The first to sixth embodiments have described examples in which the present invention is used for a dual-tube type hydraulic shock absorber, but the present invention is not limited thereto, and the present invention may be used for a mono-tube type hydraulic shock absorber in which an outer tube is eliminated and a gas chamber is formed with a slidable partition body on a side of the lower chamber **23** opposite to the upper chamber **22** in the cylinder **4**, or can be used for any shock absorber including a pressure control valve that uses a packing valve having a structure in which a seal member is provided in a disc.
(249) According to a first aspect of the embodiment described above, a shock absorber includes a cylinder in which a working fluid is sealed, a piston provided to be slidable in the cylinder and partitioning the inside of the cylinder into two chambers, a piston rod connected to the piston and extending to the outside of the cylinder, a passage through which a working fluid flows from the chamber on an upstream side to the chamber on a downstream side in the cylinder due to movement of the piston, a first damping force generation mechanism provided in the passage formed in the piston to generate a damping force, and a second damping force generation mechanism disposed separately from the first damping force generation mechanism and provided in the piston rod. The second damping force generation mechanism includes a valve seat member, a sub-valve provided in the valve seat member, and a cap member covering one end side of the second damping force generation mechanism and at least a part of an outer circumference of the valve seat member. A communication passage which allows the inside and outside of the cap member to communicate is formed at one end side of the cap member. A biasing member provided so that one end surface side is in contact with an outer circumferential side of the cap member with respect to the communication passage, and a bendable flexible disc provided so that the other end surface side of the biasing member is in contact therewith are disposed. Thereby, generation of abnormal noise can be suppressed.
(250) According to a second aspect, in the first aspect, the biasing member is provided on both one end surface side and the other end surface side of the flexible disc.
(251) According to a third aspect, in the first or second aspect, the biasing member is formed of a plate-shaped disc spring which is formed so that an outer circumferential surface side is bent to be in contact with the flexible disc and an inner circumferential surface side is bent to be in contact with a bottom part of the cap member.
(252) According to a fourth aspect, in any one of the first to third aspects, an outer circumferential end portion of the biasing member comes into contact with a cylindrical part of the cap member which covers the valve seat member.
(253) According to a fifth aspect, in any one of the first to third aspects, a guide member provided between the biasing member and the flexible disc to be in contact with an inner circumferential side of the biasing member is provided.
(254) According to a sixth aspect, in any one of the first to fifth aspects, the bottom part of the cap member includes a planar disc which is in contact with the biasing member.

INDUSTRIAL APPLICABILITY

(255) According to the shock absorber described above, generation of abnormal noise can be suppressed.

REFERENCE SIGNS LIST

(256) **1**, **1**A to **1**E Shock absorber **4** Cylinder **21** Piston **22** Upper chamber (chamber) **23** Lower chamber (chamber) Piston rod **41**, **42** First damping force generation mechanism **72**, **92** First passage (passage) **95** Cap member **100** Flexible disc **104** Spring contact disc (bottom part) **107**, **110** Sub-valve **109** Valve seat member **116**, **117** Disc spring (biasing member) **116**E, **117**E Seal disc (biasing member) **122** Bottom part **124** Cylindrical part **148**, **302** Communication passage **173**, **183** Second damping force generation mechanism **411**B, **411**C Planar disc **431**D, **432**D Guide member

## Claims

1. A shock absorber comprising: a cylinder in which a working fluid is sealed; a piston provided to be slidable in the cylinder and partitioning an inside of the cylinder into two chambers; a piston rod connected to the piston and extending to an outside of the cylinder; a passage through which a working fluid flows from the chamber on an upstream side to the chamber on a downstream side in the cylinder due to movement of the piston; a first damping force generation mechanism provided in the passage formed in the piston to generate a damping force; and a second damping force generation mechanism disposed separately from the first damping force generation mechanism and provided in the piston rod, wherein the second damping force generation mechanism includes: a valve seat member; a sub-valve provided in the valve seat member; and a cap member covering one end side of the second damping force generation mechanism and at least a part of an outer circumference of the valve seat member, and on one end side of the cap member, a communication passage which allows the inside and outside of the cap member to communicate is formed, and a biasing member provided so that one end surface side is in contact with an outer circumferential side of the cap member with respect to the communication passage, and a bendable flexible disc provided so that the other end surface of the biasing member is in contact therewith are disposed.
2. The shock absorber according to claim 1, wherein the biasing member is provided on both one end surface side and another end surface side of the flexible disc.
3. The shock absorber according to claim 1, wherein the biasing member is formed of a plate-shaped disc spring which is formed so that an outer circumferential surface side is bent to be in contact with the flexible disc and an inner circumferential surface side is bent to be in contact with a bottom part of the cap member.
4. The shock absorber according to claim 1, wherein an outer circumferential end portion of the biasing member comes into contact with a cylindrical part of the cap member which covers the valve seat member.
5. The shock absorber according to claim 1, further comprising a guide member provided between the biasing member and the flexible disc to be in contact with an inner circumferential side of the biasing member.
6. The shock absorber according to claim 1, wherein the bottom part of the cap member includes a planar disc which is in contact with the biasing member.